

Galois representations of atypical abelian varieties

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1 Introduction

sec:intro

We say that an abelian variety is atypical if its geometric endomorphism algebra is strictly larger than $\mathbb{Q}$. In this article, we study the Galois representations of geometrically simple atypical abelian varieties. Let A/F be a g -dimensional abelian variety whose geometric endomorphism algebra is a division ring D (so $D := \text{End}_{\overline{F}}^0(A)$) with degree r and center E with $e := [E : \mathbb{Q}]$. Note that A is atypical if and only if $D \neq \mathbb{Q}$. Our main tool is a structure theorem which allows us to describe the Galois representation of A in terms of Galois representations of dimension $\frac{2g}{er}$. Let $\Sigma_E := \{\iota : E \hookrightarrow \overline{\mathbb{Q}}\}$ and $E_0 := \text{End}_F^0(A) \cap E$. We write $V_\ell(A) := H_{\text{ét}}^1(A, \mathbb{Q}_\ell)^\vee$, $\rho_{A,\ell}$ for the Galois representation on $V_\ell(A)$, and $\mathcal{V}(A)$ for the compatible system of ℓ -adic representations $\{\rho_{A,\ell}\}_\ell$.

If a compatible system $\mathcal{R}'$ of Galois representations has coefficients in a field E' but all the characteristic polynomials are actually defined over $E'' \subset E'$, we say that $\mathcal{R}'$ coefficient-wise descends to E'' , and write $\text{Des}_{E''} \mathcal{R}'$ for the compatible system $\mathcal{R}'$ viewed as a compatible system with coefficients in E'' (see Section 2.4 for more details).

1.1 A special case – abelian surfaces with geometric quaternionic multiplication

We first state a special case of our theorem, and use it to deduce a new modularity result.

hm:structure-intro

Theorem 1.1. *Let $A/\mathbb{Q}$ be an abelian surface with $D := \text{End}_{\mathbb{Q}}^0(A)$ a quaternion algebra with center $\mathbb{Q}$ (so $E = E_0 = \mathbb{Q}$). Then, there exists a two-dimensional strongly irreducible compatible system $\mathcal{R}$ and a two-dimensional Artin compatible system $\mathcal{A}$ such that*

1. *The compatible system $\mathcal{R} \otimes \mathcal{A}$ coefficient-wise descends to $\mathbb{Q}$; and*
2. *$\mathcal{V}(A) \cong \text{Des}_{\mathbb{Q}}(\mathcal{R} \otimes \mathcal{A})$.*

Theorem 1.1 implies that for any ℓ , $\rho_{A,\ell} \otimes \overline{\mathbb{Q}_\ell} \cong \xi \otimes \varrho$ for a strongly irreducible two-dimensional $G_{\mathbb{Q}}$ representation ξ and a two-dimensional Artin representation ϱ . The representation ϱ cannot be a character because otherwise we would have $\text{End}_{\mathbb{Q}}^0 A = D$ – because the division ring $\text{End}_{\mathbb{Q}}^0 A$ acts faithfully on the two-dimensional $\mathbb{Q}$ -vector space $\text{Lie}(A)$, $\text{Lie}(A)$ is a free module over $\text{End}_{\mathbb{Q}}^0 A$. Because $\xi : G_{\mathbb{Q}} \rightarrow \text{GL}_2(\overline{\mathbb{Q}_\ell})$ is a geometric (unramified at all but finitely many primes, de Rham at ℓ) Galois representation with Hodge–Tate weights $(0, 1)$, it is the Galois representation of a classical modular form [Kis09, KW09] and is modular.

Because $\rho_{A,\ell}$ is self-dual up to similitude, ξ and ϱ are individually self-dual up to similitude, and one can show (Lemma 3.5) that the Weil pairing on $\rho_{A,\ell}$ is the product of pairings on ξ and ϱ . Because ξ is strongly irreducible, Schur’s lemma implies that ξ preserves a unique pairing up to scaling. Because ξ is two-dimensional, this pairing is symplectic, so the pairing on ϱ needs to be orthogonal. Therefore, ϱ is either a direct sum of characters of $G_{\mathbb{Q}}$ or the induction of a character

of G_K , where $K/\mathbb{Q}$ is a quadratic extension. In the first case, $\rho_{A,\ell}$ is reducible and $\text{End}_{\mathbb{Q}}^0(A)$ is a quadratic field, meaning that A is of GL_2 -type and is hence modular (see e.g. [Rib04, Theorem 4.4]). In the second case, there is a Galois character η of G_K such that

$$\rho_{A,\ell} \cong \xi \otimes \text{Ind}_{G_K}^{G_{\mathbb{Q}}} \eta \cong \text{Ind}_{G_K}^{G_{\mathbb{Q}}} (\text{Res}_{G_K} \xi \otimes \eta).$$

In particular, if π_f is the automorphic representation associated to ξ , then $V_{\ell}(A)$ arises as the automorphic induction of a twisted base change of π , $\text{Ind}_{K/\mathbb{Q}}(\text{BC}_{K/\mathbb{Q}}(\pi_f) \boxtimes \eta)$. In particular, A is automorphic in either of these two cases. Automorphy in the second case was proved by [?, Theorem 5.3], who associate a Siegel modular form to A . Using modern potential automorphy theorems for two-dimensional Galois representations ([Sno09, ACC⁺23], see Theorem 2.4), we deduce the following.

Theorem 1.2. *Let F be a totally real or CM number field and A/F be an abelian surface with $\text{End}_{\mathbb{Q}}^0(A) = D$, where D is a quaternion algebra. Then, there exists a two-dimensional strongly irreducible compatible system $\mathcal{R}$ and a two-dimensional Artin compatible system $\mathcal{A}$ such that*

1. *The compatible system $\mathcal{R} \otimes \mathcal{A}$ coefficient-wise descends to $\mathbb{Q}$;*
2. *$\mathcal{V}(A) \cong \text{Des}_{\mathbb{Q}}(\mathcal{R} \otimes \mathcal{A})$; and*
3. *A is potentially modular, and is modular whenever any r_{λ} in $\mathcal{R}$ is modular.*

When F is totally real, Theorem 1.2 is superseded by e.g. [BCGP21a, Theorem 10.2.1], who show potential modularity of all abelian surfaces over F .

1.2 Our main structural result

In stating Main Theorem 1, we use the term “Schur unobstructed” and the notation CRes – these will be defined properly in Section 2.4, but for now, here is a high level explanation. We say that a compatible system $\mathcal{R}' = (r'_{\lambda})_{\lambda}$ with coefficients in E' is Schur unobstructed if r'_{λ} has coefficients in E'_{λ} (and not just $\overline{E'_{\lambda}}$) for every prime λ of E' . If $\mathcal{R}'$ is Schur unobstructed and $E'' \subset E$, the “coefficient restriction” $\text{CRes}_{E'/E''}(\mathcal{R})$ is a compatible system that, when E'/E'' is Galois, can be thought of as the direct sum of the $\text{Gal}(E'/E'')$ -conjugates of $\mathcal{R}$. In particular, if $\mathcal{R}$ is an n -dimensional compatible system with coefficients in E' , $\text{CRes}_{E'/E''}(\mathcal{R})$ is a $[E' : E'']n$ -dimensional compatible system with coefficients in E'' . We say that $\mathcal{R}'$ coefficient-wise descends to a subfield E'' if the Frobenius polynomials of $\mathcal{R}'$ are actually valued in $E''[X]$, in which case we can view $\mathcal{R}'$ as a compatible system with coefficients in E'' . Recall the definitions of $E_0 \subset E \subset D$.

Main Theorem 1. *Let A be a geometrically simple abelian variety of dimension g , let L/F be the unique minimal extension over which all the geometric endomorphisms of A are defined, and let K/F be the minimal subextension of L over which $E \hookrightarrow \text{End}_K^0(A)$. Then, there exist a compatible system $\mathcal{R}$ of strongly irreducible $\frac{2g}{\text{er}}$ -dimensional G_K -representations and a compatible system $\mathcal{A}$ of r -dimensional Artin G_K -representations, well-defined up to twist by a finite-order character, such that*

1. *The projective kernel of $\mathcal{A}$ is G_L ;*
2. *$\mathcal{R} \otimes \mathcal{A}$ is a Schur unobstructed (see Section 2.4) compatible system that coefficient-wise descends to E ;*

3. $\text{Ind}_K^F \text{Des}_E(\mathcal{R} \otimes \mathcal{A})$ is a Schur unobstructed compatible system that coefficient-wise descends to E_0 ; and

4.

$$\mathcal{V}(A) \cong \text{CRes}_{E_0/\mathbb{Q}}(\text{Des}_{E_0}(\text{Ind}_K^F \text{Des}_E(\mathcal{R} \otimes \mathcal{A}))).$$

Furthermore, $\mathcal{R}$ is of symplectic type if A is geometrically type I or II, orthogonal type if A is geometrically type III, and unitary type if A is geometrically type IV.

Specializing Main Theorem 1 to the case of a single ℓ , we obtain the following corollary. Let $\Sigma_{E_0} := \text{Hom}_{\mathbb{Q}}(E_0, \overline{\mathbb{Q}})$.

Corollary 1.3. *With the notation of Main Theorem 1, for any rational prime ℓ , there are representations $\{\xi_{\iota_0}\}_{\iota_0 \in \Sigma_{E_0}}$ and $\{\varrho_{\iota_0}\}_{\iota_0 \in \Sigma_{E_0}}$ such that*

$$\rho_{A,\ell} \otimes \overline{\mathbb{Q}_\ell} \cong \bigoplus_{\iota_0 \in \Sigma_{E_0}} \text{Ind}_{G_K}^{G_F}(\xi_{\iota_0} \otimes \varrho_{\iota_0}).$$

Main Theorem 1 is not especially exciting when D is a field. When D is a division ring however, being able to pull out the dependence on r into an Artin representation enables us to understand $\mathcal{V}$ in terms of a lower-dimensional system $\mathcal{R}$. It is not hard to see how one can obtain $\frac{2g}{re}$ -dimensional direct summands in the Galois representation of A after restricting to a field L/F over which all the geometric endomorphisms of A are defined. Of particular importance here is that $\mathcal{R}$ is defined over a field K/F which in practice is much smaller than L . In particular, this K/F is the minimal subextension of L/F over which A has endomorphisms by E .

Remark. Applying Main Theorem 1 when $\text{End}^0(A) = E_0 = E$ is totally real, we recover a theorem of Fité–Florit–Guitart [FFG24, Main Theorem] in the geometrically simple case. In our language, for a geometrically simple A of geometric Albert type I, II, or III, their theorem describes $\text{Res}_{G_{K'}} \mathcal{V}(A)$, where $K'/K/F$ is an extension such that the Artin compatible system $\mathcal{A}$ (which is a representation of G_K) becomes abelian.

1.3 Atypical abelian fourfolds and potential modularity

Our main application of Main Theorem 1 is to the potential modularity of atypical abelian fourfolds. Atypical abelian fourfolds are interesting because they can exhibit all four Albert types, and more generally the simplest examples of many different algebraic cycle phenomena appear in abelian fourfolds. We prove potential modularity for many different atypical abelian fourfolds. Following the classification of Moonen–Zarhin [MZ95, Table 1], we describe our results in [?].

We say that an abelian variety A/F is geometrically type $X(e)$, $X \in \{\text{I}, \text{II}, \text{III}\}$ and $e \in \mathbb{Z}_{>0}$, if A is geometrically Albert type X and the center of $\text{End}_{\overline{F}}^0(A)$ has degree e . We say that A/F is geometrically type $\text{IV}(r, e_+)$ if it is geometrically Albert type IV and $\text{End}_{\overline{F}}^0(A)$ is a division algebra with index r whose center is a CM field whose totally real subfield has degree e_+ . Suppose that F is totally real. We say that A/F is “odd type” if every member r_λ of the compatible system $\mathcal{R}$ associated to A by Main Theorem 1 is totally odd – this means that the similitude character of r_λ is -1 on every complex conjugation of F . Loosely inspired by [BCGP21b, p. 254], we say that a geometrically type I(2) abelian fourfold A/F is “challenging” if $\mathcal{V}(A) \cong \text{Ind}_K^F \mathcal{R}$ for K a number field which is not totally real.

Main Theorem 2. *Let F be a field and A/F an abelian fourfold such that either:*

1. F is totally real and A/F is of geometric type $I(2)$ and not challenging;
2. F is totally real and A/F is odd type and of geometric type $II(1)$; or
3. F is either totally real or CM and A is of geometric type $I(4)$, $II(2)$, $IV(1,2)$, or $IV(1,4)$.

Then, A is potentially modular, and is modular if any representation in the compatible system $\mathcal{R}$ associated to A by Main Theorem 1 is modular. These results are summarized in Table 1.

Type	I(1)	I(2)		I(4)	II(1)		II(2)	III	IV(1,1)	IV(1,2)	IV(1,4)
Representation of $\mathfrak{hg}_{\mathbb{C}}$ on $H^1(A(\mathbb{C}), \mathbb{C})$	$\mathfrak{sp}_8$ or $\mathfrak{sl}_2^{\oplus 3}$	$\mathfrak{sp}_4^{\oplus 2}$		$\mathfrak{sl}_2^{\oplus 4}$	$\mathfrak{sp}_4$		$\mathfrak{sl}_2^{\oplus 2}$	$\mathfrak{so}_4$	$\mathfrak{gl}_4$ or $\mathfrak{sl}_4$	$\mathfrak{gl}_2$	$\mathbb{C}^{\oplus 8}$ or $\mathbb{C}^{\oplus 7}$
Potentially modular?	Open	Easy Yes	Challenging Open	Yes ‡	odd type Yes	Even-type Open	Yes	Open	Open	Yes ‡	Yes ‡

Table 1: A summary of the potential modularity results in this paper for abelian fourfolds over totally real fields. The icon ‡ indicates that potential modularity also holds for abelian fourfolds over CM fields.

tab:results

The most interesting case of Main Theorem 2 is that of odd type geometrically type $II(1)$ abelian fourfolds, and this is what we now discuss. Let A/F be an abelian fourfold with $D := \text{End}_{\overline{F}}^0(A)$ an indefinite quaternion algebra with center $E = \mathbb{Q}$. Corollary 1.3 implies that for any rational prime ℓ , there exist a 4-dimensional strongly irreducible symplectic representation ξ and a 2-dimensional Artin representation ϱ such that

$$V_{\ell}(A) \otimes \overline{\mathbb{Q}_{\ell}} \cong \xi \otimes \varrho. \quad (1)$$

eq:fourfold-deco

By assumption, A is odd type, so ξ is totally odd. Note that, even though ξ is only well-defined up to twist, being odd does not depend on this. Boxer–Calegari–Gee–Pilloni [BCGP21b, Theorem 8.4.1] prove a modularity lifting theorem applying to odd type geometric Galois representations $\xi: G_F \rightarrow \text{GSp}_4(\overline{\mathbb{Q}_{\ell}})$ with “big” residual image and Hodge–Tate weights $(0, 0, 1, 1)$ at every $v|\ell$. In [BCGP21b, 9.2], potential modularity for abelian surfaces is proved using a Moret-Baily argument. They work with a variety $Y/\mathbb{Q}$ whose F' -points parameterize polarized abelian surfaces B/F' along with $G_{F'}$ -module isomorphisms $B[p] \xrightarrow{\sim} A[p]$ and $B[q] \xrightarrow{\sim} \overline{\rho}_q$ for some $\overline{\rho}_q$ known to be modular. Conditional on appropriate subsets of $Y(\mathbb{R})$, $Y(\mathbb{Q}_p)$, and $Y(\mathbb{Q}_q)$ being nonempty, the Moret-Baily theorem implies that Y has a point over a totally real point F' – as usual for potential modularity arguments, two applications of the modularity lifting theorem then imply that A/F' is modular.

When A/F is an odd type abelian fourfold with geometric endomorphisms by an indefinite quaternion algebra A , we might hope to proceed in the same way, with Y replaced with the moduli of abelian fourfolds with endomorphisms containing D and again the appropriate Galois module structure on the pq -torsion. However, there is a problem – even though F is totally real, the abelian variety A does not define a real point on Y unless the geometric endomorphisms of A are also defined over a totally real field. Therefore, we instead work with a Y' of Y which – in addition to prescribing a Galois module structure on the pq -torsion, also prescribes a Galois action on the geometric endomorphism algebra. This Y' can be realized as a twist of the usual PEL Shimura variety parameterizing abelian varieties with endomorphisms by D . Once we have Y' , the rest of the Moret-Baily argument goes through in essentially the same way, and we deduce the following.

Theorem 1.4. *Let F be a totally real field and A/F be an odd type abelian fourfold with $\text{End}_{\overline{\mathbb{Q}}}^0(A) = D$, where D is an indefinite quaternion algebra with center $\mathbb{Q}$. Then, there exists a compatible system $\mathcal{R}$ of four-dimensional strongly irreducible symplectic (odd) representations and a compatible system $\mathcal{A}$ of two-dimensional Artin representations with cyclic or dihedral projective image such that*

1. $\mathcal{R} \otimes \mathcal{A}$ descends to $\mathbb{Q}$;

thm:II-1-pot-mod

2. $\mathcal{V}(A) \cong \text{Des}_{\mathbb{Q}}(\mathcal{R} \otimes \mathcal{A})$; and
3. A is potentially modular, and is modular whenever any r_{λ} in $\mathcal{R}$ is modular.

The case of geometric type I(2) ($\text{End}_{\overline{F}}^0(A) = E$, E real quadratic field) is similar. In this case, $\mathcal{R}$ is again a four-dimensional compatible system of symplectic type, but in this case it is guaranteed to be odd. The issue of finding sufficiently many ordinary primes is slightly more subtle in this setting – it is analogous to showing that a weight 2 modular form with coefficients in a quadratic field has infinitely many ordinary primes – but once this is resolved we again apply Moret-Bailly to an appropriately twisted moduli space.

Separately from the potential modularity results, we also resolve a question posed in [BCGP21b, 10.4.1]: we prove that an abelian fourfold (a so-called “fake abelian surface”) associated to a cuspidal automorphic representation of GL_4 of symplectic type will necessarily be type II, i.e. have endomorphism algebra an indefinite quaternion division algebra.

1.4 Even-type geometric Galois representations

Because Theorem 1.4 only applies to odd type abelian fourfolds, we may ask if even-type geometrically type II abelian fourfolds even exist. For such an abelian fourfold, $\mathcal{R}$ would be a four-dimensional even symplectic compatible system of geometric Galois representations. Under the usual conjectures, an automorphic form on GSp_4 associated to $\mathcal{R}$ for such abelian fourfolds would not be a Siegel modular form but rather a Siegel Maass form. Unfortunately, we do not even know how to associate Galois representations to algebraic Maass forms, and certainly do not have modularity theorems in this setting.

However, Calegari [?] showed that any two-dimensional geometric Galois representation with Zariski-dense image is odd. Because an even-type geometrically type II abelian fourfold would imply the existence of a compatible system $\mathcal{R}$ of even geometric Galois representations with Zariski-dense image in GSp_4 , one might conjecture¹ that this is impossible, like in the two-dimensional case – more generally, one might conjecture that $\mathcal{R}$ is symplectic (resp. orthogonal) if and only if it odd (resp. even). Such a conjecture would also be concordant with the results of [BLGGT14, Theorem A], who prove potential automorphy theorems (for geometric Galois representations with regular Hodge–Tate weights) exactly in the odd symplectic and even orthogonal cases.

However, this conjecture is not true! Applying Main Theorem 1, Lemma 4.4, and an argument of Frank Calegari (Lemma 5.7), we are able to show that such “even-type” abelian fourfolds exist. We are extremely grateful to F. Calegari for allowing us to include his argument in this article.

thm:even-gsp4 **Main Theorem 3.** *There exist type II abelian fourfolds A/F , F a totally real field, for which the associated four-dimensional compatible system $\mathcal{R}$ is even and symplectic-type. In particular, there exists a representation $\xi: G_F \rightarrow \text{GSp}_4(\overline{\mathbb{Q}}_{\ell})$ such that ξ is geometric, has Zariski dense image, and is totally even.*

1.5 Acknowledgements

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¹As I myself did for several months!

surrounding Lemma 2.10 and Example 2.11, and to Mateo Attanasio for helpful discussions on Section 2.6. Part of this work was completed while I was supported by an NSF Graduate Research Fellowship.

2 Preliminaries

sec:prelim

2.1 Notation and nomenclature

Given $n \in \mathbb{Z}_{\geq 1}$, we write $[n] := \{1, \dots, n\}$. We write G_F for the absolute Galois group of a field F and $|F|$ for the set of places in F . Let G be a group and V a vector space over a field k . Given a representation $\xi: G \rightarrow \mathrm{GL}(V)$ and a subgroup $H \subseteq G$, we write $\mathrm{Res}_H \xi$ for the restriction of ξ to H . We say that ξ is strongly (absolutely) irreducible if $\mathrm{Res}_H \xi$ is irreducible for every finite-index subgroup $H \subseteq G$. Given finite-dimensional characteristic 0 representations ξ and ξ' of characteristic zero of a compact group G , we write $\langle \xi, \xi' \rangle := \dim \mathrm{Hom}_{k[G]}(\xi, \xi') = \dim \mathrm{Hom}_{k[G]}(\xi', \xi)$. Given an automorphism $\tau \in \mathrm{Aut}(k)$, we write $\xi^\tau := \tau \circ \xi$. We call ξ^τ a coefficient-wise or “external” Galois conjugate of ξ . We write $\mathrm{Proj} \xi$ for the projective representation $G \xrightarrow{\xi} \mathrm{GL}(V) \rightarrow \mathrm{PGL}(V)$ associated to ξ . Let H be a normal subgroup of G , ρ be a representation of H , and σ be an element of G . We write ρ_σ for the representation of H defined by $\rho_\sigma(h) := \rho(\sigma h \sigma^{-1})$ for all $h \in H$. We call ρ_σ an “internal” Galois conjugate of ρ . Observe that $(\rho^\tau)_\sigma = (\rho_\sigma)^\tau$ for $\sigma \in G$ and $\tau \in G_k$.

We say that ξ is an extension of ρ (to G) if $\mathrm{Res}_H \xi \cong \rho$. We say that a projective representation $\xi': G \rightarrow \mathrm{PGL}(V)$ is a projective extension of ρ (to G) if $\mathrm{Res}_H \xi' \cong \mathrm{Proj} \rho$. Given a projective representation ξ' , we will sometimes abuse notation and also write ξ' for an arbitrary lift (as a function) $\xi': G \rightarrow \mathrm{GL}(V)$ – in general, this ξ' need not be a homomorphism.

When F is totally real, we say that a representation ξ of G_F that is self-dual up to similitude character ν is odd (resp. even) if ν is odd (resp. even), i.e. if $\nu(c) = -1$ for every complex conjugation $c \in G_F$. We write ε for the cyclotomic character.

Given an abelian variety A/F , we write $\mathrm{End}_F^0(A) := \mathrm{End}_F(A) \otimes_{\mathbb{Z}} \mathbb{Q}$. We call $\mathrm{End}_F^0(A)$ the geometric endomorphism algebra of A .

2.2 Embeddings

Throughout this paper, we fix embeddings $j_\ell: \overline{\mathbb{Q}} \hookrightarrow \overline{\mathbb{Q}}_\ell$ for all rational primes ℓ . This gives rise to an embedding $G_{\mathbb{Q}_\ell} \hookrightarrow G_{\mathbb{Q}}$ for every ℓ . For a number field E , we write $\Sigma_E := \mathrm{Hom}_{\mathbb{Q}}(E, \overline{\mathbb{Q}})$. There is a transitive action of $G_{\mathbb{Q}}$ on Σ_E by postcomposition. For each rational prime ℓ , j_ℓ gives an embedding $G_{\mathbb{Q}_\ell} \hookrightarrow G_{\mathbb{Q}}$, and the orbits of Σ_E under the resulting $G_{\mathbb{Q}_\ell}$ -action correspond to primes λ of E dividing ℓ . An embedded number field is a tuple (E, ι_0) consisting of a number field E and a marked embedding $\iota_0 \in \Sigma_E$. In the context of an embedded number field, for each $\iota \in \Sigma_E$, we write $\tilde{\iota}: \overline{\mathbb{Q}} \rightarrow \overline{\mathbb{Q}}$ for an element of $G_{\mathbb{Q}}$ such that $\tilde{\iota} \circ \iota_0 = \iota \in \Sigma_E$. For any $\tau \in G_{\mathbb{Q}}$, $\tau \iota_0 \in \Sigma_E$ is the embedding satisfying $\tilde{\tau} \iota_0 := \tilde{\iota} \circ \tau \circ \iota_0 \in \Sigma_E$. The embedded number field (E', ι'_0) is an extension of (E, ι_0) if $E' \supset E$ and ι'_0 extends ι_0 , and we call $(E', \iota'_0)/(E, \iota_0)$ an extension of embedded number fields.

2.3 Automorphy

subsec:aut

Recall that an ℓ -adic representation ξ of G_K is said to be an automorphic Galois representation if there is an automorphic form over K whose L -function is the same as that of ξ . We say that ξ is a potentially automorphic Galois representation if there exists a finite extension K'/K such that $\mathrm{Res}_{G_{K'}} \xi$ is automorphic.

An abelian variety is (potentially) automorphic if any of its ℓ -adic representations are (potentially) automorphic. Given a (potentially) automorphic Galois representation, it is useful to understand operations which preserve (potential) automorphy.

lem:aut-props **Lemma 2.1.** *Let $\xi: G_K \rightarrow \mathrm{GL}_n(\overline{\mathbb{Q}_\ell})$ be a (potentially) automorphic Galois representation.*

1. *For any $r \in \mathbb{Z}_{\geq 1}$, $\xi^{\oplus r}$ is (potentially) automorphic;*
2. *Any twist of ξ by a character is (potentially) automorphic;*
3. *If K/F is a Galois extension, then for any $\sigma \in \mathrm{Gal}(K/F)$, any conjugate ξ_σ of ξ is (potentially) automorphic;*

Proof. Let L be an extension of K such that $\mathrm{Res}_{G_L} \xi$ is automorphic.

1. The direct sum of automorphic Galois representations is automorphic (via the isobaric sum). The analogue for potential automorphy follows because $\mathrm{Res}_{G_L}(\xi^{\oplus r}) \cong (\mathrm{Res}_{G_L} \xi)^{\oplus r}$;
2. This follows from class field theory;
3. Let π be the automorphic representation associated to $\mathrm{Res}_{G_L} \xi$ for some extension L/F . Then, $\pi \circ \sigma$ is an automorphic representation of $\mathrm{GL}_n(\mathbb{A}_{\sigma^{-1}(L)})$ whose Satake parameters at unramified primes are exactly the Frobenius characteristic polynomials of $\mathrm{Res}_{G_{\sigma^{-1}(L)}} \xi_\sigma$.

□

It does not follow from Lemma 2.1 that the direct sum of two different potentially automorphic Galois representations is itself potentially automorphic. Let ξ_1 and ξ_2 be representations such that $\mathrm{Res}_{L_1} \xi_1$ and $\mathrm{Res}_{L_2} \xi_2$ automorphic. Unless $L_1 L_2 / L_1$ and $L_1 L_2 / L_2$ both happen to be solvable, we cannot apply base change to deduce that $\mathrm{Res}_{L_1 L_2} \xi_1$ and $\mathrm{Res}_{L_1 L_2} \xi_2$ are automorphic.

However, in practice, potential automorphy of a G_K -representation ξ not arise in a vacuum. Typically, we have two ingredients. The first is a modularity lifting theorem which – for any finite extension L/K satisfying some conditions which are preserved under compositum and Galois closure – given a G_L -representation ρ' satisfying some p -adic Hodge theory conditions whose residual representation is automorphic and has big image, implies that ρ' itself is automorphic. The second is a residual representation $\bar{\rho}: G_K \rightarrow \mathrm{GL}_m(\overline{\mathbb{F}_\ell})$ that is strongly automorphic, i.e. $\mathrm{Res}_{G_L} \bar{\rho}$ is automorphic for every L/K , and satisfies the aforementioned big image conditions for L/K linearly disjoint from some extension K^{avoid}/K . Then, an argument due to Taylor [Tay02] is used to show that an L -point on an appropriate moduli space – for an L satisfying all the above conditions – implies that $\mathrm{Res}_{G_L} \xi$ is automorphic.

The Moret–Bailly theorem is then used to find an L -point – for an L satisfying all the above conditions – on an appropriate moduli space Y , and any such point implies the automorphy of $\mathrm{Res}_{G_L} \xi$. If Y has an L -point, it certainly has a $\tilde{L}$ -point for any $\tilde{L} \supset L$. However, such a $\tilde{L}$ need not satisfy the hypotheses of our modularity lifting theorem and our strongly automorphic residual representation with big image.

To ameliorate this, we introduce the following notion.

def:strong-pot-aut **Definition 2.2.** A Galois representation $\xi: G_K \rightarrow \mathrm{GL}_n(\overline{\mathbb{Q}_\ell})$ is **MB-potentially automorphic** if there exist

- a smooth geometrically connected variety Y/K ;
- a field K^{avoid}/K ;

- a finite set of places S of K ;
- nonempty open subsets $\Omega_v \subset Y_{K_v}$

such that for any point $P \in Y(L)$ – where L/K is Galois, L/K is disjoint from K^{avoid}/K , and every $v \in S$ splits completely in L/K – such that $P \in \Omega_v$ for all v , $\text{Res}_{G_L} \xi$ is modular.

The point of Definition 2.2 is that the Moret–Bailly theorem applied to Y with the additional data guarantees the existence of such a P . However, given a finite collection representations $\{\xi_i\}$ which are all MB-potentially automorphic with associated data $\{Y_i\}$, $\{K_i^{\text{avoid}}\}$, $\{S_i\}$, and $\{\Omega_{v,i}\}$, we can apply Moret–Bailly to $\prod_i Y_i$, taking $S := \bigcup_i S_i$, K^{avoid} to be the compositum of the $\{K_i^{\text{avoid}}\}$, and $\Omega_v := \prod_i \Omega_{v,i}$ for $v \in S$. This gives us a field extension L over which all of the $\{\xi_i\}$ are simultaneously automorphic.

All of the automorphy lifting theorems we make use of in this paper produce MB-potentially automorphic representations.

Lemma 2.3. *Let $\xi, \xi': G_K \rightarrow \text{GL}_n(\overline{\mathbb{Q}_\ell})$ be (MB-potentially) automorphic Galois representations.*

1. *If ξ and ξ' are (MB-potentially) automorphic, then $\xi \oplus \xi'$ is (MB-potentially) automorphic;*
2. *If K/K'' is a Galois extension, $\text{Ind}_{G_K}^{G_{K''}} \xi$ is potentially automorphic.*
3. *If K'/K is a solvable Galois extension, $\text{Res}_{G_{K'}} \xi$ is (MB-potentially) automorphic;*

Proof. 1. If ξ_1 and ξ_2 are MB-potentially automorphic, then $\xi_1 \oplus \xi_2$ is MB-potentially automorphic with respect to $Y := Y_1 \times Y_2$, as discussed after Definition 2.2.

2. Because ξ is MB-potentially automorphic, we may assume that there is a Galois extension L/K such that $\text{Res}_{G_L} \xi$ is automorphic. Therefore, for any $\sigma \in \text{Gal}(L/K)$, $(\text{Res}_{G_L} \xi)_\sigma$ is still a representation of G_L . We have

$$\text{Res}_{G_L} \left(\text{Res}_{G_K} \text{Ind}_{G_K}^{G_{K''}} \xi \right) = \text{Res}_{G_L} \left(\bigoplus_{\sigma \in \text{Gal}(K/K'')} \xi_\sigma \right) = \bigoplus_{\sigma \in \text{Gal}(K/K'')} (\text{Res}_{G_L} \xi)_\sigma,$$

so the result follows from Lemma 2.1.

3. Because ξ is MB-potentially automorphic, we may assume that there is an extension L/K linearly disjoint from K'/K such that $\text{Res}_{G_L} \xi$ is automorphic. Then, LK'/L is a solvable Galois extension, and by solvable base change [AC89], $\text{Res}_{G_{LK'}} \text{Res}_{G_L} \xi = \text{Res}_{G_{LK'}} \text{Res}_{G_{K'}} \xi$ is automorphic.

□

The following two theorems are our main tools for proving potential automorphy results. The first is a potential automorphy theorem for two-dimensional representations over totally real or CM fields. The second is a modularity lifting theorem for representations valued in $\text{GSp}_4(\overline{\mathbb{Q}_p})$

Theorem 2.4 ([ACC⁺23, Sno09]). *Let $K/\mathbb{Q}$ be a totally real or CM extension, p be a prime split in $K/\mathbb{Q}$, and $\xi: G_K \rightarrow \text{GL}_2(\overline{\mathbb{Q}_p})$ a continuous irreducible representation such that*

1. *ξ is unramified at all but finitely many primes of F ;*
2. *$\xi|_{G_{F_v}}$ is de Rham with Hodge–Tate weights $(0, 1)$ for every $v|p$*

3. There exists a finite field $k/\mathbb{F}_p$ such that the image of the residual representation of ξ contains $\mathrm{SL}_2(k)$.

Then, ξ is potentially automorphic.

Theorem 2.5 ([BCGP21b, Theorem 8.4.1]). Let F' be a totally real number field and $\xi: G_{F'} \rightarrow \mathrm{GSp}_4(\overline{\mathbb{Q}_p})$ a continuous Galois representation such that

- ξ has cyclotomic similitude character;
- For all $v|p$, $\xi|_{G_{F'_v}}$ is p -distinguished weight 2 ordinary;
- $\bar{\xi}$ is vast and tidy in the sense of [BCGP21b, Definitions 7.5.6 and 7.5.11];
- $\bar{\xi}$ is automorphic, and the associated automorphic form π has an associated (characteristic 0) Galois representation ρ_π ;
- ξ and ρ_π are pure at all finite places of F' ;
- The restriction $\xi|_{G_{F'_v}}$ is potentially crystalline and has Hodge–Tate weights $(0, 0, 1, 1)$ for every $v|p$

Then, ξ is automorphic.

Note that $\bar{\xi}$ is vast and tidy if its image contains $\mathrm{Sp}_4(\overline{\mathbb{F}_p})$.

2.4 Compatible systems of Galois representations

Definition 2.6. Let K be a number field and (E, ι_0) an embedded number field and S a finite set of primes of K . A weakly compatible system of m -dimensional Galois representations over a field K unramified away from S with coefficients in (E, ι_0) consists of, for each rational prime ℓ and each $\iota \in \Sigma_E$, a representation $r_{\ell, \iota}: G_K \rightarrow \mathrm{GL}_m(\overline{\mathbb{Q}_\ell})$ such that:

1. Away from S and the primes of K dividing ℓ , $r_{\ell, \iota}$ is unramified;
2. If ι and ι' are in the same $G_{\mathbb{Q}_\ell}$ -orbit (see Section 2.2), then $r_{\iota, \ell}$ and $r_{\iota', \ell}$ are externally Galois conjugate by an automorphism of $G_{\mathbb{Q}_\ell}$;
3. There is a polynomial $Q_v(X) \in \mathrm{im}(\iota_0)[X] \subset \overline{\mathbb{Q}}[X]$ (independent of ℓ) such that the characteristic polynomial of $r_{\ell, \iota}(\mathrm{Frob}_v)$ is $(j_\ell \circ \tilde{\iota})(Q_v)$;
4. Each $r_{\ell, \iota}$ is de Rham at places $v|\ell$ of K , and crystalline when $v \notin S$.

We denote these data by the 5-tuple $\mathcal{R} = (K; (E, \iota_0), S, (Q_v(X))_v, (r_\lambda)_\lambda)$.

By the second property in ??, we can specify $\mathcal{R}$ by specifying, for each ℓ , one $r_{\ell, \iota}$ in each $G_{\mathbb{Q}_\ell}$ -orbit. Because the $G_{\mathbb{Q}_\ell}$ -orbits in Σ_E correspond to prime ideals of E above ℓ , this is equivalent to specifying representations $(r_\lambda)_\lambda$ for each $\lambda|\ell$. Typical treatments of compatible systems (e.g. [BLGGT14, 5.1]) implicitly fix once and for all, for each ℓ , a collection of orbit representatives $\{\iota_\lambda\}_{\lambda|\ell}$, and define compatible systems by specifying representations $(r_\lambda)_\lambda$ where λ ranges over the primes of E . In our setting, explicitly tracking all $[E : \mathbb{Q}]$ representations for each ℓ will ameliorate the definition of certain operations involving Galois conjugation, but does not actually change the underlying objects which we are working with.

Remark. We deviate from the usual notation for compatible systems (e.g. [BLGGT14, 5.1]), by specifying the base field K and by not specifying the Hodge–Tate weights $\{H_\tau\}_{\tau: K \hookrightarrow \overline{E}}$. We specify these extra data because we work with operations on compatible systems that change K (e.g. induction) and that are sensitive to the choices of embeddings (e.g. “coefficient-wise” Galois conjugation). While the Hodge–Tate weights are important, the majority of the compatible systems appearing in this article are self-dual up to similitude and are associated to abelian varieties, and the Hodge–Tate weights are always half zeroes and half ones in these cases. If $\mathcal{R}$ is unitary type, then this need not hold, but in this article we do not work substantially with unitary type compatible systems.

We now define several operations on weakly compatible systems.

Definition 2.7. Let $\mathcal{R} := (K; (E, \iota_0), S, \{Q_v(X)\}_v, \{r_{\ell, \iota}\}_{\ell, \iota})$ be a weakly compatible system.

1. For any extension of embedded fields $(E', \iota'_0)/(E, \iota_0)$, we can define the “coefficient-wise extension”

$$\mathcal{R}_{(E', \iota'_0)} := (K; (E', \iota'_0), S_{E'}, (Q_v(X))_v, (r_{\lambda'} := r_{\lambda'} \cap E)_{\lambda'})$$

2. Given another compatible system $\mathcal{R}' = (K'; E', S', (Q'_v(X))_v, (r'_{\lambda'})_{\lambda'})$ with $(E, \iota_0) = (E', \iota'_0)$, we can define

$$\mathcal{R} \oplus \mathcal{R}' := (K; (E, \iota_0), S \cup S', \{Q_v Q'_v\}_v, \{r_{\ell, \iota} \oplus r'_{\ell, \iota}\}_{\ell, \iota})$$

If $E \neq E'$, we define $\mathcal{R} \oplus \mathcal{R}' := \mathcal{R}_{EE'} \oplus \mathcal{R}'_{EE'}$.

3. The tensor product of compatible systems $\mathcal{R}$ and $\mathcal{R}'$ is defined analogously to the direct sum, with $\otimes$ replacing $\oplus$ everywhere.
4. For any $\sigma \in G_{\mathbb{Q}}$, the “internal” Galois conjugate (in the sense of Section 2.1) of $\mathcal{R}$ by σ is

$$\mathcal{R}_\sigma := (\sigma(K); E, \sigma(S), \{Q_{\sigma(v)}(X)\}_v, \{(r_{\ell, \iota})_\sigma\}_{\ell, \iota}).$$

5. For any $\tau \in G_{\mathbb{Q}}$, the “external” or “coefficient-wise” Galois conjugate is

$${}^\tau \mathcal{R} := (K; \tau(E), S, \{{}^\tau Q_v(X)\}_v, \{r'_{\ell, \iota} := r_{\ell, \tau \iota}\}_{\ell, \iota}).$$

6. For any (E'', ι''_0) contained in (E, ι_0) , suppose that $Q_v(X) \in \text{im}(\iota''_0)[X]$ for every prime $v \notin S$ of K . We define

$$\text{Des}_{E''}(\mathcal{R}) := (K; (E'', S, \{Q_v(X)\}_v, \{r_{\ell, \iota''} := r_{\ell, \iota} \text{ for any } \iota \in \Sigma_E \text{ extending } \iota''\}_{\ell, \iota''})).$$

Because we are working with characteristic 0 representations of a compact group, the Brauer–Nesbitt theorem ensures that this operation is well-defined (independent of the choice of ι extending each ι''). If the hypothesis for applying $\text{Des}_{E''}$ holds, we say that $\mathcal{R}$ coefficient-wise descends to E'' .

7. For any $(E'', \iota''_0) \subset (E, \iota_0)$,

$$\text{CRes}_{E/E''}(\mathcal{R}) := \text{Des}_{E''} \left(\bigoplus_{\tau \in \mathcal{T}} \mathcal{R}^\tau \right),$$

where $\mathcal{T} \subset G_{\mathbb{Q}}$ is any collection of $[E : E'']$ elements such that $\{\tau \circ \iota_0\}_{\tau \in \mathcal{T}} \subset \Sigma_E$ is the set of embeddings of E extending ι''_0 . In particular, if $E'' = \mathbb{Q}$, then in the notation of Section 2.2, we can take $\mathcal{T} := \{\tilde{\iota} : \iota \in \Sigma_E\}$.

8. For any subfield $F \subset K$,

$$\text{Ind}_K^F(\mathcal{R}) := \left(F; E, S' := \{v|_F : v \in S\} \cup \{v|\text{disc}(K/F)\}, (Q'_{v'}(X))_{v' \in |F| \setminus S'}, (\text{Ind}_{G_K}^{G_F} r_\lambda)_\lambda \right), (\iota_\lambda)_\lambda,$$

where $Q'_v(X)$ is an appropriate polynomial which one can check has coefficients in $E[X]$ (or see e.g. [?, VII §10]).

9. For any extension K'/K , we define

$$\text{Res}_{K'}(\mathcal{R}) := \left(K'; E, S' := \{w : w|v \text{ for some } v \in S\}, (Q_w(X))_{w \in |K'| \setminus S'}, (\text{Res}_{G_{K'}} r_\lambda)_\lambda, (\iota_\lambda)_\lambda \right),$$

where, writing f_w for the residue field degree at w , each $Q_w(X) \in E[X]$ is the polynomial whose roots are the f_w^{th} powers of the roots of $Q_v(X)$.

Because the hypothesis for applying $\text{Des}_{E''}$ is a condition on Frobenius characteristic polynomials, and the Frobenius characteristic polynomials in a compatible system are “independent of (ι, ℓ) ” in an appropriate sense (the third condition of Definition 2.6), we can verify this hypothesis on a single $r_{\iota, \ell}$. We record this as a lemma for future use.

Lemma 2.8. *To prove coefficient-wise descent of $\mathcal{R}$ to (E'', ι''_0) , it suffices to fix a rational prime ℓ and $\iota \in \Sigma_E$ and show that for every $v \notin S$, $r_{\iota, \ell}(\text{Frob}_v)$ has characteristic polynomial in $(j_\ell \circ \tilde{\iota} \circ \iota''_0)(E'') = (j_\ell \circ \iota'')(E'')$, where $\iota'' := \tilde{\iota} \circ \iota''_0 \in \Sigma_{E''}$.*

Note that this is a substantially stronger condition than simply showing that the characteristic polynomials lie in $E''_{\lambda''}$ for some $\lambda''|\lambda$ (e.g. the latter is automatic when λ'' splits completely in E/E'').

The main motivation to define compatible systems using embedded coefficient fields and, for each ℓ , to track $\{r_{\ell, \iota}\}_{\iota \in \Sigma_E}$ instead of simply the $G_{\mathbb{Q}_\ell}$ -orbit representatives $\{r_\lambda\}_{\lambda|\ell}$, is in order to define the operation $\text{CRes}_{E/E''}$ above.

Example 2.9. Let A be the GL_2 -type abelian variety A_f associated to a weight 2 modular newform f with coefficients in $E/\mathbb{Q}$. Let $\mathcal{R}_f$ be the 2-dimensional compatible system of $G_{\mathbb{Q}}$ -representations with coefficients in E attached to f . Then,

$$\mathcal{V}(A) \cong \text{CRes}_{E/\mathbb{Q}} \mathcal{R}_f.$$

The point of our machinery is that this is a stronger result than we would be able to state while working with a single ℓ -adic representation. For simplicity, assume that $E/\mathbb{Q}$ is Galois. For any ℓ , we have

$$\rho_{A, \ell} \otimes \overline{\mathbb{Q}_\ell} \cong \bigoplus_{\tau \in \text{Gal}(E/\mathbb{Q})} \rho_{\tau f}, \quad (2) \quad \text{\texttt{\{eq:gl2-decomp\}}}$$

where each $\rho_{\tau f} : G_{\mathbb{Q}} \rightarrow \text{GL}_2(\overline{\mathbb{Q}_\ell})$ is the 2-dimensional Galois representation associated to the newform ([DS05, Theorem 6.5.4]) ${}^\tau f$. We would like to think of $\rho_{A, \ell}$ as the direct sum of the “Galois conjugates” of ρ_f . If ℓ is totally inert in E , this is literally correct, as $\text{Gal}(E/\mathbb{Q}) \cong \text{Gal}(E_\lambda/\mathbb{Q}_\ell)$ and we have $\rho_{\tau f} \cong {}^\tau \rho_f$ and $\rho_{A, \ell}$ is the direct sum of external Galois conjugates of ρ_f . However, if ℓ splits completely in E , the representation ρ_f are valued in $\mathbb{Q}_\ell$ and has no nontrivial Galois conjugates. In this case, the Galois representations associated to ${}^\tau f$ cannot be obtained from ρ_f , and must be logged separately, and in order to reconstruct Equation (2), we need the full data of the compatible system.

If $\mathcal{R}$ and $\mathcal{R}'$ are absolutely irreducible, then $\text{CRes}_{E/E''} \mathcal{R} \cong \text{CRes}_{E/E''} \mathcal{R}'$ if and only if $\mathcal{R}' \cong \mathcal{R}^\tau$ for some $\tau \in G_{\mathbb{Q}}$.

We say that $\mathcal{R}$ is Schur unobstructed if for all ℓ and ι , $r_{\ell,\iota}$ stabilizes an m -dimensional E_λ -vector space – where $\lambda|\ell$ is the prime ideal of E associated to ι – and hence can be defined over E_λ (and not just over $\overline{E}_\lambda = \overline{\mathbb{Q}_\ell}$).

Remark. Note that $\mathcal{R}$ coefficient-wise descending to E'' is only a statement about the field of definition of the Frobenius characteristic polynomials. Each representation $r_{\ell,\iota''}$ will still typically only be defined over $\overline{\mathbb{Q}_\ell}$. In particular, $\mathcal{R}$ coefficient-wise descending to E'' does not mean that $\text{Des}_{E''} \mathcal{R}$ is Schur unobstructed.

We say that $\mathcal{R}$ is of symplectic (resp. orthogonal) type if every r_λ is essentially self-dual and preserves a symplectic (resp. orthogonal) pairing up to similitude. If the base field of $\mathcal{R}$ is totally real and $\mathcal{R}$ is symplectic or orthogonal type, we write that $\mathcal{R}$ is odd type (resp. even-type) if its similitude character is totally odd (resp. totally even). Suppose that E is CM, with totally real subfield E^+ and complex conjugation automorphism c . We say that r_λ is essentially conjugate self-dual $r_\lambda^\vee \cong^c r_{c(\lambda)} \otimes \chi_\lambda$ for some character χ_λ – here, $r_{c(\lambda)}^c$ denotes the coefficient-wise application of c to $r_{c(\lambda)}$. We say that $\mathcal{R}$ is of unitary type if every r_λ is We say that $\mathcal{R}$ is of unitary type if every r_λ is essentially conjugate self-dual.

2.5 Representation theoretic lemmas

For ξ a representation of G , $\text{Res}_H \xi \cong (\text{Res}_H \xi)_\sigma$ for all $\sigma \in G$. The following lemma captures the extent to which the converse (or at least, some projective version thereof) holds.

Lemma 2.10. *Let G be a group, $H \subseteq G$ be a finite-index normal subgroup, and V a vector space over an algebraically closed field $\bar{k}$. Let $\rho: H \rightarrow \text{GL}(V)$ be an irreducible representation such that $\rho_\sigma \cong \rho$ for all $\sigma \in G$.*

We can associate a cocycle $[c] \in H^2(G/H, \bar{k}^\times)$ to (G, ρ) , and there exists a representation $\tilde{\rho}: G \rightarrow \text{GL}(V)$ such that $\text{Res}_H \tilde{\rho} \cong \rho$ (an extension of ρ to G) if and only if $[c]$ vanishes.

Example 2.11. The nontrivial character of $\{\pm 1\} \subset Q_8$ is stable under conjugation, but is not the restriction of any character of Q_8 . I learned this example from Chengyang Bao. Indeed, $Q_8/\{\pm 1\} \cong (\mathbb{Z}/2\mathbb{Z})^2$ and one can check that $H^2((\mathbb{Z}/2\mathbb{Z})^2, \mathbb{C}^\times) \cong \mathbb{Z}/2\mathbb{Z}$.

Proof. This is a standard result in Clifford theory, and a proof can also be found in e.g. [Isa06]. Choose representatives $\tilde{\sigma} \in G$ for each $\sigma \in G/H$. To specify an extension $\tilde{\rho}$ to G , it is sufficient to specify the values of $T_{\tilde{\sigma}} := \tilde{\rho}(\tilde{\sigma})$ for all $\sigma \in G/H$. We can then define $\tilde{\rho}(\tilde{\sigma}h) := T_{\tilde{\sigma}}\rho(h)$. Let $h_{\sigma,\sigma'} := \widetilde{\sigma\sigma'}^{-1} \tilde{\sigma}\tilde{\sigma}'$. The map $\tilde{\rho}$ is a group homomorphism if and only if

$$T_{\tilde{\sigma}}\rho(h)T_{\tilde{\sigma}'}\rho(h') = T_{\widetilde{\sigma\sigma'}}\rho(h_{\sigma,\sigma'})\rho(\tilde{\sigma}'^{-1}h\tilde{\sigma}'h')$$

for all $\sigma, \sigma' \in G/H$ and $h, h' \in H$. Equivalently, we require that

$$T_{\tilde{\sigma}}T_{\tilde{\sigma}'} = T_{\widetilde{\sigma\sigma'}}\rho(h_{\sigma,\sigma'}) \quad \text{and} \quad \rho(\tilde{\sigma}'^{-1}h\tilde{\sigma}') = T_{\tilde{\sigma}'}^{-1}\rho(h)T_{\tilde{\sigma}'} \quad (3) \quad \{\text{eq:clifford-1}\}$$

for all $\sigma, \sigma' \in G/H$ and $h, h' \in H$. The second condition in Equation (3) means that each $T_{\tilde{\sigma}}$ must intertwine ρ and $\rho_{\tilde{\sigma}}$ – such intertwiners exist by our assumption that $\rho \cong \rho_{\tilde{\sigma}}$, and are well-defined up to scaling by Schur's lemma. By definition of $h_{\sigma,\sigma'}$,

$$\rho(x) = \rho((h_{\sigma,\sigma'}^{-1} \widetilde{\sigma\sigma'}^{-1} \tilde{\sigma}\tilde{\sigma}')x(h_{\sigma,\sigma'}^{-1} \widetilde{\sigma\sigma'}^{-1} \tilde{\sigma}\tilde{\sigma}')^{-1}) = (\rho(h_{\sigma,\sigma'}^{-1})T_{\widetilde{\sigma\sigma'}}^{-1}T_{\tilde{\sigma}}^{-1}T_{\tilde{\sigma}'}\rho(x)(\rho(h_{\sigma,\sigma'}^{-1})T_{\widetilde{\sigma\sigma'}}^{-1}T_{\tilde{\sigma}}^{-1}T_{\tilde{\sigma}'}^{-1})^{-1}.$$

Again by Schur's lemma, $\rho(h_{\sigma,\sigma'}^{-1})T_{\sigma\sigma'}^{-1}T_{\tilde{\sigma}}T_{\tilde{\sigma}'}$ is a scalar, so the first condition in Equation (3) always holds up to scaling. Let $c(\sigma,\sigma') := \rho(h_{\sigma,\sigma'})^{-1}T_{\sigma\sigma'}^{-1}T_{\tilde{\sigma}}T_{\tilde{\sigma}'}$. This quantity is independent of the choices of lifts $\tilde{\sigma}$ and $\tilde{\sigma}'$. Furthermore, one can check that c is a 2-cocycle for G/H , and that the associated class $[c] \in H^2(G/H, \bar{k}^\times)$ is well-defined independent of the choices of intertwiners $\{T_{\tilde{\sigma}}\}$. In particular, $[c]$ vanishes if and only if there exist intertwiners $\{T_{\tilde{\sigma}}\}$ satisfying the conditions of Equation (3). $\square$

Remark. Suppose we want to work with projective representations instead of genuine representations. Suppose that $\text{Proj } \rho_\sigma \cong \text{Proj } \rho$ for all $\sigma \in G$. If we are only interested in finding projective extensions of ρ , then we only need Equation (3) to hold up to scaling, and setting $\tilde{\rho}(\tilde{\sigma}) := T_{\tilde{\sigma}}$ (for any choice of intertwiner) for all σ is good enough for this. Therefore, there always exists a projective representation ξ such that $\text{Res}_H \tilde{\rho} \cong \rho$ (with no cocycle condition).

In general, we will not be able to assume that $H^2(G/H, \bar{k}^\times) = 0$, although this is true when G/H is cyclic. In our application however, G is the absolute Galois group of a number field, and we can use Tate's theorem to say something stronger.

Corollary 2.12. *Let F'/F be a finite Galois extension of number fields, $V/\bar{k}$ be a finite-dimensional vector space over an algebraically closed field, and $\rho: G_{F'} \rightarrow \text{GL}(V)$ be an irreducible representation such that $\rho_\sigma \cong \rho$ for all $\sigma \in \text{Gal}(F'/F)$. Then, there exists a representation $\xi: G_F \rightarrow \text{GL}(V)$ and a finite-order character $\psi: G_{F'} \rightarrow \bar{k}^\times$ such that $\text{Res}_{G_{F'}} \xi \cong \rho \otimes \psi$, and $\text{Res}_{\ker \psi} \xi \cong \text{Res}_{\ker \psi} \rho$.*

Proof. Tate proved (see [Pat19, Theorem 2.11]) that $H^2(G_F, \mathbb{Q}/\mathbb{Z}) = 0$. Since $\text{coker}(\mu_\infty(\bar{k}) \hookrightarrow \bar{k}^\times)$ is uniquely divisible, $H^2(G_F, \text{coker}(\mu_\infty(\bar{k}) \hookrightarrow \bar{k}^\times)) = 0$ implies that $H^2(G_F, \bar{k}^\times) = 0$. Because

$$H^2(G_F, \bar{k}^\times) = H^2(\varprojlim \text{Gal}(\tilde{F}/F), \bar{k}^\times) = \varinjlim_{\tilde{F}/F} H^2(\text{Gal}(\tilde{F}/F), \bar{k}^\times),$$

for any nonzero class $[c] \in H^2(\text{Gal}(F'/F), \bar{k}^\times)$ there exists a finite Galois extension $\tilde{F}/F'$ such that the image of $[c]$ in $H^2(\text{Gal}(\tilde{F}/F), \bar{k}^\times)$ is 0. By Lemma 2.10, it follows that $\text{Res}_{G_{\tilde{F}}} \rho$ admits an extension ξ to G . Because ρ is irreducible, it follows from Frobenius reciprocity there is a finite-order character ψ of $\text{Gal}(\tilde{F}/F')$ such that $\text{Res}_{G_{F'}} \xi \cong \rho \otimes \psi$. $\square$

Lemma 2.13. *Let H be a finite-index normal subgroup of G . Suppose that ρ is a representation of H and $\tilde{\rho}$ a representation of G extending ρ . If ξ is a representation of G such that $\langle \text{Res}_H \xi, \rho \rangle = r$, then there exists an r -dimensional representation ϱ of G/H such that $\tilde{\rho} \otimes \varrho$ is a subrepresentation of ξ .*

Proof. Because ρ has a lift to G ,

$$\text{Ind}_H^G \rho = \bigoplus_{\varrho'} (\tilde{\rho} \otimes \varrho')^{\oplus \dim \varrho'},$$

where ϱ' ranges over the irreducible representations of G/H . By Frobenius reciprocity,

$$r = \langle \text{Res}_H \xi, \rho \rangle = \langle \xi, \bigoplus_{\varrho'} (\tilde{\rho} \otimes \varrho')^{\oplus \dim \varrho'} \rangle.$$

As such, $\tilde{\rho} \otimes \left(\bigoplus_{\varrho' \in S} \varrho' \right)$ is a subrepresentation of ξ for some collection of ϱ' such that $\sum_{\varrho' \in S} \dim \varrho' = r$, and we define $\varrho := \bigoplus_{\varrho' \in S} \varrho'$. $\square$

2.6 Twisted moduli spaces of abelian schemes

sec:twisted-moduli

Let D be an endomorphism algebra of index r and center E of degree e equipped with a positive involution $\dagger$, and let $\mathcal{O} \subset D$ be a $\mathbb{Z}$ -lattice stable under $\dagger$.

Let V be a $2g$ -dimensional vector space over $\mathbb{Q}$ and $\Lambda \subset V$ a lattice defining a weight 1 $\mathcal{O}$ -Hodge structure with polarization ψ such that $\psi(dx, y) = \psi(x, d^\dagger y)$. Let $\phi: \Lambda \times \Lambda \rightarrow \mathbb{Z}$ be a D -skew Hermitian pairing on $\Lambda \otimes_{\mathbb{Z}} \mathbb{R}$ such that $\text{tr}_{D/\mathbb{Q}} \phi = \psi \otimes_{\mathbb{Z}} \mathbb{R}$. Let h be the Hodge cocharacter associated to Λ .

Define $G := \text{CSp}_D(V, \phi)$. Because $(B, \dagger)$ is of type (C) in the language of [Mil05, Chapter 8], [Zin83, 3.1] (see also [Mil05, Proposition 8.14]) implies that there exists a unique conjugacy class X of homomorphisms $\text{Res}_{\mathbb{C}/\mathbb{R}} \mathbb{G}_m \rightarrow G_{\mathbb{R}}$ such that (G, X) is a PEL Shimura datum. Let F_0 be the reflex field of the PEL Shimura datum (G, X) . Fix an extension F'/F_0 and a finite Galois extension L'/F' , and a finite set of rational primes S , and choose homomorphisms

$$\rho_{\text{End}}: \text{Gal}(L'/F') \longrightarrow \text{Aut}_{\mathbb{Q}}(D) \quad \text{and} \quad \bar{\rho}_{\ell}: \text{Gal}(L'/F') \longrightarrow \text{GL}_{2g}(\mathbb{F}_{\ell}) \text{ for all } \ell \in S.$$

Let n be the product of the primes in S . We can combine the data of the $\bar{\rho}_{\ell}$ into a single group homomorphism $\bar{\rho}_n: \text{Gal}(L'/F') \rightarrow \text{GL}_2(\mathbb{Z}/n\mathbb{Z})$. Let G'_n denote the étale group scheme over F' (of order n^2) associated to $\bar{\rho}_n$. Let $K \subset G(\mathbb{A}_f)$ be the compact open subgroup associated to the principal level- n structure.

The following definitions are largely motivated by [Lan13, Definition 1.4.1.2]. Consider the moduli functor $\mathcal{M}: \mathbf{Sch}_{/F'} \rightarrow \mathbf{Set}$ sending a scheme S to the set of quadruples $(A, \lambda, \iota, \eta)$ (to be defined momentarily) up to isomorphism (also to be defined momentarily).

For any scheme S/F' , we define $S_{L'} := S \times_{\text{Spec } F'} \text{Spec } L'$. In each quadruple:

- A is an abelian scheme over S of relative dimension g ;
- $\lambda: A \rightarrow A^\vee$ is a $\mathbb{Q}^\times$ -polarization (in the sense of [Lan13, Definition 1.3.2.19]) of A
- $\iota: D \hookrightarrow \text{End}^0(A \times_S S_{L'})$ defines an $\mathcal{O}$ -structure of $(A \times_S S_{L'}, \lambda)$ (in the sense of [Lan13, Definition 1.3.3.1]) such that the action of $\text{Gal}(L'/F')$ on $\iota(D) \subseteq \text{End}^0(A \times_S S_{L'})$ is given by ρ_D ;
- $\eta: (G'_n)_S \xrightarrow{\sim} A[n]$ is an isomorphism of group schemes such the induced isomorphism $\eta_{S_{L'}}: (\mathbb{Z}/n\mathbb{Z})_S \xrightarrow{\sim} A[n] \times_S S_{L'}$ is an (integral) principal level- n structure of (A, λ, ι) of type $(\Lambda \otimes \hat{Z}, \psi)$.
- $\text{Lie}_{A \times_S S_{L'}}$ satisfies the Kottwitz determinant condition described in [Lan13, Definition 1.3.4.1] given by $(\Lambda \otimes_{\mathbb{Z}} \mathbb{R}, \psi, h)$.

An isomorphism $(A, \lambda, \iota, \eta) \rightarrow (A', \lambda', \iota', \eta')$ is given by an isomorphism $f: A \xrightarrow{\sim} A'$ of abelian schemes such that

1. $\lambda = f^\vee \circ \lambda' \circ f$
2. For any $d \in \mathcal{O}$, the map

$$\begin{aligned} f_*: \text{End}^0(A \times_S S_{L'}) &\longrightarrow \text{End}^0(A' \times_S S_{L'}) \\ \iota(d) &\longmapsto f \circ \iota(d) \circ f^{-1} \end{aligned}$$

is an isomorphism of $\text{Gal}(L'/F')$ -modules that sends $\iota(d)$ to $\iota'(d)$.

3. $f|_{A[n]} \circ \eta = \eta'$.

Theorem 2.14. *Let $\mathcal{M}$ be the moduli functor described above. There is a scheme Y'/F representing $\mathcal{M}$ such that*

1. Y' is a twist of $\mathrm{Sh}_K(G, X)_F$ – concretely, $Y'_{L'}$ and $\mathrm{Sh}_K(G, X)_{L'}$ are isomorphic.
2. Y' is smooth and geometrically connected if and only if $\mathrm{Sh}_K(G, X)_{L'}$ is, and in particular when K is small enough and G is connected

Proof. Let $Y := \mathrm{Sh}_K(G, X)$. We will realize Y' via Galois descent from $Y_{L'}$. To do this, we must define a descent datum, i.e. a collection $(f_\sigma: {}^\sigma Y_{L'} \rightarrow Y_{L'})_{\sigma \in \mathrm{Gal}(L'/F')}$ of morphisms of L' -schemes satisfying the cocycle condition $f_{\sigma\tau} = f_\sigma \circ {}^\sigma f_\tau$ for all $\sigma, \tau \in G$. This is equivalent to giving morphisms of F' -schemes $(f'_\sigma: Y_{L'} \rightarrow Y_{L'})_{\sigma \in \mathrm{Gal}(L'/F')}$ such that

$$\begin{array}{ccc} Y_{L'} & \xrightarrow{f'_\sigma} & Y_{L'} \\ \downarrow & & \downarrow \\ \mathrm{Spec} L' & \xrightarrow{\sigma^{-1}} & \mathrm{Spec} L' \end{array}$$

and satisfying the condition $f'_{\sigma\tau} = f'_\sigma \circ f'_\tau$ – here, $\mathrm{Spec} L \xrightarrow{\sigma^{-1}} \mathrm{Spec} L$ is the map of schemes corresponding to the homomorphism $L \xleftarrow{\sigma^{-1}} L$. The equivalence between the two formulations is captured by the following diagram:

$$\begin{array}{ccccc} & & f'_\sigma & & \\ & \nearrow & & \searrow & \\ Y_{L'} & \xrightarrow{\sigma^{-1}} & {}^\sigma Y_{L'} & \xrightarrow{f_\sigma} & Y_{L'} \\ \downarrow & & \downarrow & & \downarrow \\ \mathrm{Spec} L' & \xrightarrow{\sigma^{-1}} & \mathrm{Spec} L' & \xrightarrow{\mathrm{id}} & \mathrm{Spec} L' \\ & \searrow & & \nearrow & \\ & & \sigma^{-1} & & \end{array}$$

By Yoneda's lemma, we can define each map f'_σ by specifying $f'_\sigma(S): Y_{L'}({}^{\sigma^{-1}}S) \rightarrow Y_{L'}(S)$ which satisfy the composition law and are compatible with morphisms of L' -schemes $S \rightarrow S'$.

If we were simply trying to descend $Y_{L'}$ to Y , the “usual” descent datum $(g_\sigma)_{\sigma \in \mathrm{Gal}(L'/F')}$ would be given by

$$\begin{aligned} g'_\sigma: \quad & Y({}^{\sigma^{-1}}S) \longrightarrow Y(S) \\ & ({}^{\sigma^{-1}}A, {}^{\sigma^{-1}}\lambda, {}^{\sigma^{-1}}\iota, {}^{\sigma^{-1}}\circ\eta) \longmapsto (A, \lambda, \iota, \eta) \end{aligned}$$

Given representations ρ_{End} and $\bar{\rho}_n$, we can define the maps

$$\begin{aligned} h_\sigma: \quad & Y(S) \longrightarrow Y(S) \\ & (A, \lambda, \iota, \eta) \longmapsto (A, \lambda, \iota \circ \rho_{\mathrm{End}}(\sigma)^{-1}, \eta \circ \bar{\rho}_n(\sigma)^{-1}). \end{aligned}$$

The map h_σ can be thought of as relabelling the endomorphism and level structure on each abelian scheme parameterized by $Y_{L'}$ according to $\rho_D(\sigma)$ and $\rho_n(\sigma)$. Since ρ_D and ρ_n are homomorphisms, the $(h_\sigma)_\sigma$ satisfy the condition $h_\sigma h_\tau = h_{\sigma\tau}$. Because the $(h_\sigma)_\sigma$ commute with the g'_σ (because

precomposition commutes with postcomposition), we can define a new Galois descent datum $(f'_\sigma)_\sigma$ by $f'_\sigma := h_\sigma \circ g'_\sigma$. We write $Y'/\text{Spec } F$ for the resulting variety over F .

If $\text{Sh}_K(G, X)_{L'}$ is not smooth or geometrically connected then Y' will not be. Conversely, if $\text{Sh}_K(G, X)_{L'}$ is smooth and geometrically connected, then Y' will be as well because Galois (more generally, fpqc) descent preserves these properties (see e.g. [Poo17, Table 1]).

For K neat (and in particular, sufficiently small), $\text{Sh}_K(G, X)_{L'}$ will be smooth. $\text{Sh}_K(G, X)$ is geometrically connected if and only if G^{der} is simply connected. $\square$

3 A structure theorem for Galois representations of abelian varieties

3.1 Proving the structure theorem

Let A/F be a g -dimensional geometrically simple abelian variety over a number field F . Let $D := \text{End}_{\overline{F}}^0(A)$ (the geometric endomorphism algebra of A), and let E be the center of D . E is a number field, and we define $e := [E : \mathbb{Q}]$ and $r := \sqrt{\dim_E D}$. By [Sil92, Proposition 2.2], there exists a unique minimal extension L/F such that all the endomorphisms in D are defined over L , and this L/F is Galois. Let $\mathcal{V} := \mathcal{V}(A) = (F; \mathbb{Q}, S, (Q_v(X))_v, (\rho_{A,\ell})_\ell)$ be the compatible system of Galois representations associated to A , where $V_\ell(A) := H_{\text{et}}^1(A, \mathbb{Q}_\ell)^\vee$, $\rho_{A,\ell}$ is the Galois representation on $V_\ell(A)$, and S is the set of primes of bad reduction for A .

By Faltings' theorem [Fal83], $\text{End}_{G_{F'}}(V_\ell(A)) \cong \text{End}_{F'}(A) \otimes \mathbb{Q}_\ell$ for any extension F'/F . In particular, taking $F' = L$, $\text{End}_L^0(A) \otimes \overline{\mathbb{Q}_\ell} = D \otimes \overline{\mathbb{Q}_\ell} \cong M_r(\overline{\mathbb{Q}_\ell})^e$, and so

$$\text{Res}_{G_L} V_\ell(A) \otimes \overline{\mathbb{Q}_\ell} \cong \bigoplus_{i \in [e]} \rho_i^{\oplus r} \quad (4) \quad \{\text{eq:1}\}$$

for irreducible $\frac{2g}{er}$ -dimensional representations $\{\rho_i\}_{i \in [e]}$.

By the Albert classification, $\text{End}_{F'}^0(A)$ for any F'/F is either a totally real field ($A_{F'}$ is type I), a totally indefinite quaternion algebra over a totally real field ($A_{F'}$ is type II), a totally definite quaternion algebra over a totally real field ($A_{F'}$ is type III), or a division ring whose center is a CM field ($A_{F'}$ is type IV). We say that A is geometrically type X if $A_{\overline{F}}$ has Albert type X. Note that the geometric Albert type and the Albert type can often differ, so the distinction is important.

Main Theorem 1. *Let A be a geometrically simple abelian variety of dimension g , let L/F be the unique minimal extension over which all the geometric endomorphisms of A are defined, and let K/F be the minimal subextension of L over which $E \hookrightarrow \text{End}_K^0(A)$. Then, there exist a compatible system $\mathcal{R}$ of strongly irreducible $\frac{2g}{er}$ -dimensional G_K -representations and a compatible system $\mathcal{A}$ of r -dimensional Artin G_K -representations, well-defined up to twist by a finite-order character, such that*

1. *The projective kernel of $\mathcal{A}$ is G_L ;*
2. *$\mathcal{R} \otimes \mathcal{A}$ is a Schur unobstructed (see Section 2.4) compatible system that coefficient-wise descends to E ;*
3. *$\text{Ind}_K^F \text{Des}_E(\mathcal{R} \otimes \mathcal{A})$ is a Schur unobstructed compatible system that coefficient-wise descends to E_0 ; and*

4.

$$\mathcal{V}(A) \cong \text{CRes}_{E_0/\mathbb{Q}} \left(\text{Des}_{E_0} \left(\text{Ind}_K^F \text{Des}_E(\mathcal{R} \otimes \mathcal{A}) \right) \right).$$

Furthermore, $\mathcal{R}$ is of symplectic type if A is geometrically type I or II, orthogonal type if A is geometrically type III, and unitary type if A is geometrically type IV.

Because any automorphism of D preserves its center E , the Galois action of $\text{Gal}(L/F)$ on D restricts to an action on E . The group $\text{Gal}(L/K)$ is the kernel of this action, so K/F is automatically Galois.

Suppose that $E' \hookrightarrow \text{End}_{F'}^0(A)$. Then, setting $e' := [E' : \mathbb{Q}]$, we can write

$$\text{Res}_{G_{F'}} \rho_A : G_{F'} \longrightarrow \text{GL}_{\frac{2g}{e'}}(E' \otimes \mathbb{Q}_\ell) \cong \prod_{\lambda|\ell} \text{GL}_{\frac{2g}{e'}}(E'_{\lambda'}), \quad (5) \quad \text{\texttt{eq:structure-2}}$$

and we define $\rho'_{\lambda'} : G_{F'} \rightarrow \text{GL}_{\frac{2g}{e'}}(E'_{\lambda'})$ to be the projection of $\text{Res}_{G_{F'}} \rho_A$ onto the factor corresponding to λ' . Let $V := V_\ell(A)$. Because E' is abelian, the action of E' on $V \otimes \overline{\mathbb{Q}_\ell}$ decomposes as a sum of characters $E' \rightarrow \overline{\mathbb{Q}_\ell}$ – the possible characters are indexed by $\{j_\ell \circ \iota' : \iota' \in \Sigma_{E'}\}$, and we call the space where E' acts by $j \circ \iota'$ the ι' -eigenspace $V_{\iota'}$ of $V \otimes \overline{\mathbb{Q}_\ell}$. Since the E' -action commutes with the $G_{F'}$ -action, $V_{\iota'}$ is a $G_{F'}$ -subrepresentation for all $\iota' \in \Sigma_{E'}$. Viewed in this light, the subrepresentation $\rho'_{\lambda'}$ acts on subspace where the action of E' factors through the map $E' \hookrightarrow E'_{\lambda'}$, and, over $\overline{\mathbb{Q}_\ell}$, this subspace is the direct sum of the eigenspaces $V_{\iota'}$ associated to the embeddings $j \circ \iota'$ extending $E' \hookrightarrow E'_{\lambda'}$. With S as above, let $S_{F'}$ denote the primes of F' dividing the primes of S .

prop:e-compat **Proposition 3.1** (see [Rib76, Theorem 2.1.2]). *The representations $\rho'_{\lambda'}$ appearing in Equation (5) fit into a $\frac{2g}{e'}$ -dimensional Schur unobstructed compatible system $\mathcal{W}' := (F'; E', S_{F'}, \{Q_v(X)\}, \{\rho'_{\lambda'}\})$ such that $\text{Res}_{G_{F'}} \mathcal{V} \cong \text{CRes}_{E'/\mathbb{Q}} \mathcal{W}'$.*

For any ℓ , we have $D \otimes_{\mathbb{Q}} \overline{\mathbb{Q}_\ell} \cong \prod_{\iota \in \Sigma_E} M_r(\overline{\mathbb{Q}_\ell})$. In the ι^{th} factor of this decomposition, an endomorphism $e \in E$ is represented by the scalar $(j \circ \iota)(e) \in \overline{\mathbb{Q}_\ell}$. Applying Proposition 3.1 with $F' = L$ and $E' = E$, there is a compatible system $\mathcal{U}' = (L; E, S_L, (Q_w(X))_w, (u'_\lambda)_\lambda)$, w ranging over primes of L and λ ranging over primes of E , such that $\text{Res}_{G_L} \mathcal{V} = \text{CRes}_{E/\mathbb{Q}} \mathcal{U}'$. For our choice of ℓ ,

$$\text{Res}_{G_L} \rho_A \cong \bigoplus_{\lambda|\ell} u'_\lambda, \quad (6) \quad \text{\texttt{eq:1}}$$

and there exist representations u_λ such that $u'_\lambda \otimes_{E_\lambda} \overline{\mathbb{Q}_\ell} \cong u_\lambda^{\oplus r}$, where in the tensor product we choose an arbitrary embedding of E_λ into $\overline{\mathbb{Q}_\ell}$. Note that when λ is split in D , we do not need to tensor with $\overline{\mathbb{Q}_\ell}$ in order to define u_λ . The representations u_λ are strongly irreducible as otherwise, by Falting's theorem, the geometric endomorphism algebra of A would be larger than D . For any prime w of L not in S_L , because the characteristic polynomial $u'_\lambda(\text{Frob}_w)$ is monic, has coefficients in E , and is independent of λ , the same is true for the characteristic polynomial of $u_\lambda(\text{Frob}_w)$. Therefore, $\mathcal{U} := (L; S_L, E, (Q_w(X)^{1/r})_w, (u_\lambda)_\lambda)$ also defines a (Schur obstructed, when $r > 1$) weakly compatible system with coefficients in E , and $\mathcal{U}' \cong \mathcal{U}^{\oplus r}$. Because the $(u_\lambda)_\lambda$ are strongly irreducible, $\mathcal{U}$ is strongly irreducible.

Decomposing Equation (6) even further, we have

$$\text{Res}_{G_L} \rho_A \otimes \overline{\mathbb{Q}_\ell} \cong \bigoplus_{\iota \in \Sigma_E} \xi_\iota^{\oplus r}, \quad (7) \quad \text{\texttt{eq:main-3}}$$

where $\xi_\iota^{\oplus r}$ acts on the subspace V_ι defined prior to Proposition 3.1. Consider the inner conjugation action of $\sigma \in G_F$ on the direct summands in Equation (7). We have

$$\bigoplus_{\iota \in \Sigma_E} (\xi_\iota)_\sigma^{\oplus r} \cong \text{Res}_{G_L} (\rho_A)_\sigma \cong \text{Res}_{G_L} \rho_A \cong \bigoplus_{\iota \in \Sigma_E} \xi_\iota^{\oplus r}, \quad (8) \quad \text{\texttt{eq:2}}$$

where in the penultimate isomorphism we use that ρ_A is a representation of G_F (and not just of G_L). By Equation (8), G_F acts on the set $\{\xi_\ell\}$ by inner conjugation. Because $E \hookrightarrow \text{End}_K^0(A)$, any $\sigma \in G_K$ must respect the E -eigenspace decomposition of $V_\ell(A)$ and hence must stabilize every ξ_ℓ . (As we will see later, G_K is in fact the shared stabilizer of all of the ξ_ℓ under the inner conjugation action).

Applying Proposition 3.1 with $F' = K$ and $E' = E$, there exists a compatible system of G_K -representations $\mathcal{W}$ with coefficients in E such that $\text{Res}_{G_K} \mathcal{V} \cong \text{CRes}_{E/\mathbb{Q}} \mathcal{W}$ and $\text{Res}_{G_L} \mathcal{W} \cong \mathcal{U}^{\oplus r}$. By definition, $\mathcal{U}_\sigma \cong \mathcal{U}$ for every $\sigma \in G_K$. We claim that there are compatible systems $\mathcal{R}$ and $\mathcal{A}$ over K , with coefficients in some extension E'/E , such that the compatible system $\mathcal{R} \otimes \mathcal{A}$ descends to E and $\mathcal{W} \cong \text{Des}_E(\mathcal{R} \otimes \mathcal{A})$.

For any such $\mathcal{R}$, we must have $\text{Res}_{G_L} \mathcal{R} \cong \mathcal{U}$. Fix a prime $\lambda_0 | \ell_0$ of E . We have $(u_{\lambda_0})_\sigma \cong u_{\lambda_0}$ for every $\sigma \in G_K$, so by Corollary 2.12, there exists a G_K -representation r_0 and a finite Galois extension M_0/K such that $\text{Res}_{G_{M_0}} r_0 \cong \text{Res}_{G_{M_0}} u_{\lambda_0}$. By Lemma 2.13, there then exists an Artin representation $a_0: G_K \twoheadrightarrow \text{Gal}(M_0/K) \rightarrow \text{GL}_r(\overline{\mathbb{Q}_{\ell_0}})$ with projective kernel G_L such that $w_{\lambda_0} \cong r_0 \otimes a_0$. In particular, $r_0 \otimes a_0$ is a subrepresentation of $\text{Res}_{G_K} \rho_A$.

Consider the abelian variety $A' := \text{Res}_{M_0/K} A_{M_0}$, the Weil restriction to K of the base change to M_0 of A . Let ρ_A be the ℓ_0 -adic representation of associated to A . The ℓ_0 -adic Galois representation $\rho_{A'}$ associated to A' is $\text{Ind}_{G_{M_0}}^{G_K} (\text{Res}_{G_{M_0}} \rho_A) \cong \text{Res}_{G_K} \rho_A \otimes \mathbb{Q}_{\ell_0}[\text{Gal}(M_0/K)]$. Because $a_0^\vee$ is a subrepresentation of the regular representation $\mathbb{Q}_{\ell_0}[\text{Gal}(M_0/K)]$, $a_0 \otimes a_0^\vee$ has a trivial representation summand, r_0 is a direct summand of $\rho_{A'}$. By [Gu25, Lemma 1.6], it follows that r_0 is a member of a compatible system of Galois representations $\mathcal{R}$ over K with coefficients in some extension E'/E . Concretely, this means that $r_0 = r_{\lambda'_0}$ for some prime $\lambda'_0 | \ell_0$ of E' . Up to replacing $\mathcal{R}$ with a coefficient-wise Galois conjugate, we can assume that $\lambda'_0 | \lambda_0$. The compatible system $\mathcal{R}$ is strongly irreducible because $\text{Res}_{G_L} \mathcal{R} = \mathcal{U}$ and $\mathcal{U}$ is strongly irreducible.

Assume without loss of generality that E' is large enough so that a_0 can be defined over E' . Then, the Artin representation a_0 fits into a compatible system $\mathcal{A} = \left((j_\ell \circ \sigma_{\lambda'} \circ j_{\ell_0}^{-1})(a_0) \right)_{\lambda'}$, where $\lambda' | \ell$ ranges over the primes of E' and $\sigma_{\lambda'} \in G_{\mathbb{Q}}$ is an element taking $\iota_{\lambda'_0}$ to $\iota_{\lambda'}$ (recall that these are both elements of $\Sigma_{E'}$, see Section 2.1). In particular, $a_0 = a_{\lambda'_0}$.

It follows that $\mathcal{W}' := \mathcal{R} \otimes \mathcal{A}$ is a compatible system over K with coefficients in E' . We want to show that $\mathcal{W}'$ descends to E . Fixing a prime λ' of E' , it suffices to show that the Frobenius characteristic polynomials of $w_{\lambda'} = r_{\lambda'} \otimes a_{\lambda'}$ lie in $((j_\ell \circ \iota_{\lambda'})(E))[X]$. Because $\mathcal{W}'$ is a compatible system, these characteristic polynomials are equal to $j_\ell \circ \iota_{\lambda'} \circ \iota_{\lambda'_0}^{-1} \circ j_{\ell_0}^{-1}$ applied to the Frobenius characteristic polynomial of $r_{\lambda'_0} \otimes a_{\lambda'_0} \cong w_{\lambda_0}$. As w_{λ_0} is a member of a compatible system $\mathcal{W}$ with coefficients in E and $\lambda'_0 | \lambda_0 | \ell$, its Frobenius characteristic polynomials lie in $((j_{\ell_0} \circ \iota_{\lambda_0})(E))[X]$. Because $\iota_{\lambda_0}(E) = \iota_{\lambda'_0}(E)$, it follows that $\mathcal{W}'$ descends to E and that $\mathcal{W} \cong \text{Des}_E \mathcal{W}' \cong \text{Des}_E(\mathcal{R} \otimes \mathcal{A})$ as desired.

So far, we have shown that

$$\text{Res}_{G_K} \mathcal{V} \cong \text{CRes}_{E/\mathbb{Q}} \mathcal{W} \cong \text{CRes}_{E/\mathbb{Q}} \text{Des}_E(\mathcal{R} \otimes \mathcal{A}). \quad (9) \quad \text{\textcolor{blue}{\code{eq:structure-4}}}$$

It remains to understand $\mathcal{V}$ itself.

Earlier, we fixed an ℓ and considered the inner conjugation action of $\sigma \in G_F$ on $\bigoplus_{\iota \in \Sigma_E} \xi_\ell^{\oplus r}$, and observed that G_K is contained in the intersection of the stabilizers of the ξ_ℓ . Now, we can do better. By Equation (9), we have

$$(\text{Res}_{G_K} \mathcal{V})_{(E^{\text{gal}}, \iota_0^{\text{gal}})} \cong \bigoplus_{\tau \in \mathcal{T}} \mathcal{W}^\tau.$$

For any $\sigma \in G_F$, $(\text{Res}_{G_K} \mathcal{V})_\sigma \cong \text{Res}_{G_K} \mathcal{V}$. Suppose that σ stabilizes $\mathcal{W}^\tau$ for some $\tau \in \mathcal{T}$. Because inner and outer Galois conjugation commute, it follows that σ stabilizes $\mathcal{W}^\tau$ for every $\tau \in \mathcal{T}$. But then, looking at any r_λ , we see that σ stabilizes the E -eigenspace decomposition, and hence that it lies in the kernel of the map $G_F \rightarrow \text{Aut}_{\mathbb{Q}}(E)$. In particular, any $\sigma \in G_F$ fixing any of the $\mathcal{W}^\tau$ fixes all of them and hence lies in G_K . Therefore, $\text{Gal}(K/F)$ acts freely on the $\{\mathcal{W}^\tau\}_{\tau \in \mathcal{T}}$, and hence – picking an ℓ , on the $\{w_\iota\}_{\iota \in \Sigma_E}$, where $w_\iota = \xi_\iota \otimes \varrho_\iota$. Let $S \subset \Sigma_E$ be a set of orbit representatives for the $\text{Gal}(K/F)$ action. Because $\text{Res}_{G_K} \rho_A \cong \bigoplus_{\iota \in \Sigma_E} w_\iota$ and the $\{w_\iota\}$ are pairwise nonisomorphic, Frobenius reciprocity implies that

$$\rho_A \otimes \overline{\mathbb{Q}_\ell} \cong \bigoplus_{\iota \in S} \text{Ind}_{G_K}^{G_F} w_\iota \cong \bigoplus_{\iota \in S} \text{Ind}_{G_K}^{G_F} (\xi_\iota \otimes \varrho_\iota),$$

proving Corollary 1.3. The $\text{Ind}_{G_K}^{G_F} w_\iota$ are not guaranteed to be irreducible (because w_ι need not be, as ϱ_ι can be reducible) but each $\text{Ind}_{G_K}^{G_F} w_\iota$ is a minimal direct sum of E -eigenspaces which are stable under the G_F -action (on $E \hookrightarrow \text{End}_F^0(A)$) by construction, each is necessarily an E_0 -eigenspace. Applying Proposition 3.1 with $E' = E_0$ and $F' = F$, we deduce that $\text{Ind}_K^F \mathcal{W}$ is Schur unobstructed and descends to E_0 , and furthermore that

$$\mathcal{V} \cong \text{CRes}_{E_0/\mathbb{Q}} \text{Des}_{E_0} (\text{Ind}_K^F \mathcal{W}) \cong \text{CRes}_{E_0/\mathbb{Q}} \text{Des}_{E_0} \text{Ind}_K^F \text{Des}_E (\mathcal{R} \otimes \mathcal{A}),$$

concluding the proof of Main Theorem 1.

3.2 Potential automorphy

The following is our main tool to prove potential automorphy.

Corollary 3.2. *An abelian variety A/F is (MB-potentially) automorphic if L/F is solvable and the compatible system $\mathcal{R}$ associated to A by Main Theorem 1 is (MB-potentially) automorphic.*

Proof. It suffices to show that $\rho_{A,\ell}$ is (MB-potentially) automorphic. Because $\mathcal{R}$ is (strongly potentially) automorphic, we may assume that all of the $\{\xi_{\iota_0}\}_{\iota_0}$ appearing in Corollary 1.3 are (strongly potentially) automorphic.

First, L/K is solvable, so $\text{Res}_{G_L} \xi_j$ is (strongly potentially) automorphic by Lemma 2.3. Therefore, $\text{Res}_{G_L} (\xi_j \otimes \varrho_j) \cong (\text{Res}_{G_L} \xi_j)^{\oplus r}$ is (strongly potentially) automorphic, meaning that $\xi_j \otimes \varrho_j$ itself is. Second, because K/F is solvable, applying Lemma 2.3 implies that $\text{Ind}_{G_K}^{G_F} (\xi_j \otimes \varrho_j)$ is (strongly potentially) automorphic. The direct sum of (strongly potentially) automorphic representations is (strongly potentially) automorphic, so the result follows from Corollary 1.3. $\square$

Combining Corollary 3.2, class field theory, Theorem 2.4, and noting that a two-dimensional Galois representation without big image is (after restricting to a finite extension) a direct sum of characters, we deduce the following.

Corollary 3.3. *If F is a totally real or CM field and A/F is an abelian variety whose associated compatible system $\mathcal{R}$ given by Main Theorem 1 is one- or two-dimensional, then A is potentially modular.*

3.3 Pairings

Throughout this subsection, we fix a rational prime ℓ . As noted in the proof of Main Theorem 1, $E \hookrightarrow \text{End}_K^0(A)$, and

$$\text{Res}_K V_\ell(A) \otimes \overline{\mathbb{Q}_\ell} \cong \bigoplus_{\iota \in \Sigma_E} (\xi_\iota \otimes \varrho_\iota). \quad (10)$$

lem:self-dual **Lemma 3.4.**

1. If A is not type IV, the compatible system $\text{Des}_E \mathcal{R} \otimes \mathcal{A}$ associated to A by Main Theorem 1 is of symplectic type with cyclotomic similitude character.
2. If A is type IV, the compatible system $\text{Des}_E \mathcal{R} \otimes \mathcal{A}$ associated to A by Main Theorem 1 is of unitary type with cyclotomic similitude character.

Proof. Since $V_\ell(A)^\vee \cong V_\ell(A) \otimes \varepsilon$, for any ι we have $(\xi_\iota \otimes \varrho_\iota)^\vee \cong (\xi_{\iota'} \otimes \varrho_{\iota'}) \otimes \varepsilon$ for some ι' . Let $\phi: V_\ell(A) \rightarrow V_\ell(A) \otimes \varepsilon$ be induced by the polarization on A . By the definition of the Rosati involution, for any $d \in \text{End}^0(A) \hookrightarrow \text{End}_{G_F} V_\ell(A)$ we have $d^\vee \circ \phi = \phi \circ d^\dagger$. If $e \in E \subset D$ is self-adjoint with respect to the Rosati involution (the adjoint involution for the polarization), then distinct E -eigenspaces (the $\{\xi_j \otimes \varrho_j\}$) are orthogonal with respect to the pairing ψ on $V_\ell(A)$ induced by ϕ . If A is not geometrically type IV, then this applies to every $e \in E$. The pairing on $V_\ell(A)$ is then the sum of the pairings on the E -eigenspaces, so each E -eigenspace must preserve a symplectic pairing and the first part of the result follows. If E is geometrically type IV, then we can apply the same reasoning to the eigenspaces of the totally real subfield E^+ of E . Each E^+ -eigenspace is a direct sum of two complex-conjugate E -eigenspaces corresponding to embeddings $\iota, \bar{\iota} \in \text{Hom}_{E^+}(E, \overline{\mathbb{Q}})$. Without loss of generality, suppose that x and y are in the E -eigenspace corresponding to ι . Then, $\iota(e)\psi(x, y) = \psi(ex, y) = \psi(x, e^\dagger y) = \iota(e^\dagger)\psi(x, y)$. Taking $e \in E \setminus E^+$, $\iota(e) \neq \iota(e^\dagger)$ and $\psi(x, y) = 0$. By nondegeneracy of the Weil pairing, we deduce that the two E -eigenspaces are paired against one another and that $\xi_\iota \otimes \varrho_\iota$ is essentially conjugate self-dual. Because the pairing on entire E^+ -eigenspace is alternating with cyclotomic similitude character, the pairing on each E -eigenspace is skew-Hermitian and with cyclotomic similitude character. $\square$

Let M/K be the extension through which the $\{\varrho_\iota\}$ (or equivalently, the compatible system $\mathcal{A}$) factor.

lem:xi-self-dual **Lemma 3.5.** When A is not type IV, the $\{\xi_\iota\}$ and $\{\varrho_\iota\}$ appearing in Equation (10)) (and hence, the $\{\xi_{\iota_0}\}$ and $\{\varrho_{\iota_0}\}$ appearing in Corollary 1.3) are self-dual up to similitude. Furthermore, the Weil pairing on $V_\ell(A)$ is the sum of the products of the pairings on the $\{\xi_j\}$ and $\{\varrho_j\}$.

Proof. Because $\text{Res}_{G_M} V_\ell(A) \cong (\text{Res}_{G_M} \xi)^\oplus r$, $\text{Res}_{G_M} \xi$ is self-dual up to cyclotomic similitude because $\text{Res}_{G_M} V_\ell(A)$ is. By Frobenius reciprocity and Lemma 2.13, it follows that $\xi^\vee \cong \xi \otimes (\varepsilon\nu)$, where ν is a character of $\text{Gal}(M/K)$.

Fix an $\iota_\ell := j_\ell \circ \iota: E \hookrightarrow \overline{\mathbb{Q}_\ell}$. By Lemma 3.4, $\xi_\iota \otimes \varrho_\iota$ preserves a symplectic pairing up to cyclotomic similitude. Therefore,

$$\wedge^2(\xi_\iota \otimes \varrho_\iota) \cong \text{Sym}^2 \xi_\iota \otimes \wedge^2 \varrho_\iota \oplus \wedge^2 \xi_\iota \otimes \text{Sym}^2 \varrho_\iota \quad (11)$$

must have a one-dimensional irreducible constituent. Restricting Equation (11) to G_M , at least one of $\text{Sym}^2(\text{Res}_{G_M} \xi_\iota)$ and $\wedge^2(\text{Res}_{G_M} \xi_\iota)$ has a one-dimensional irreducible constituent, and it follows that $\text{Res}_{G_M} \xi_\iota$ is essentially self-dual. By Frobenius reciprocity, ξ_ι itself is essentially self-dual. It follows that at least one of $\wedge^2 \varrho_\iota$ or $\text{Sym}^2 \varrho_\iota$ has a one-dimensional irreducible constituent, so ϱ_ι is also essentially self-dual.

Because ξ_ι is strongly irreducible, the dimension of the space of pairings preserved (up to similitude) by ξ_ι is at most one-dimensional by Schur's lemma. By Equation (11), the symplectic pairing on $\xi_\iota \otimes \varrho_\iota$ is then the product of the unique pairing on ξ and the complementary pairing on ϱ , and the (cyclotomic) similitude character of the pairing on $\xi_\iota \otimes \varrho_\iota$ is the product of the respective similitude characters. $\square$

The $\{\xi_\iota\}$ do not need to preserve symplectic pairings when $r > 1$ – as we will see, either summand in Equation (11) can have the one-dimensional subrepresentation, depending on the geometric Albert type of the abelian variety. Suppose A is of geometric Albert type II or III, so $D := \text{End}_F^0(A)$ is a quaternion algebra and $r = 2$. Let $\mathfrak{hg}_\ell$ be the Lie algebra of the ℓ -adic algebraic monodromy group (the Zariski closure over $\mathbb{Q}$ of the image of the ℓ -adic Galois representation).

Let $W_\iota := \text{Res}_M \xi_\iota$. Restricting both sides of Equation (10) to G_M , we have

$$\text{Res}_{G_M} V_\ell(A) \otimes \overline{\mathbb{Q}_\ell} \cong \bigoplus_{\iota: E \hookrightarrow \overline{\mathbb{Q}_\ell}} W_\iota^{\oplus 2}$$

as $\mathfrak{hg}_\ell$ -representations. The following result is well-known.

lem: def-orth **Lemma 3.6.** *Suppose that A is geometrically type II or III. For all $\iota: E \hookrightarrow \overline{\mathbb{Q}_\ell}$, the representation of $\mathfrak{hg}_\ell$ on W_ι is self-dual. This representation preserves a symplectic pairing if D is definite, and an orthogonal pairing if D is indefinite.*

Proof. Because $V_\iota := W_\iota^{\oplus 2} \cong \text{Res}_M(\xi_\iota \otimes \varrho_\iota) \cong (\text{Res}_M \xi_\iota)^{\oplus 2}$, self-duality of W_ι follows from the self-duality (up to similitude) of ξ_ι given by Lemma 3.5. Henceforth, we write $V := V_\iota$ and $W := W_\iota$. Because W is irreducible, the pairing is unique by Schur’s lemma. Therefore, we can write the matrix of the pairing on V as $\Omega' \otimes \Omega_W$ where Ω_W is the matrix of the pairing on W and $\Omega' = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in M_2(\overline{\mathbb{Q}_\ell})$. If $(\Omega')^T = \Omega'$, then the pairing on W is alternating, and if $(\Omega')^T = -\Omega'$, then

the pairing on W is symmetric. In the former case, we may change basis so that $\Omega' = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$, and

in the latter, $\Omega' = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$. This lets us express the pairing $\langle \cdot, \cdot \rangle_V$ on V in terms of that on W .

Since we are working with a single ι -eigenspace, we pick a corresponding embedding of D into $M_2(\overline{\mathbb{Q}_\ell})$. Because the endomorphisms in D commute with the action of $\mathfrak{hg}_\ell$, D acts via this 2×2 matrix on the decomposition $V \cong W \oplus W$. The Rosati involution respects the pairing, so writing $x, y \in V$ as $(x_1, x_2), (y_1, y_2) \in W \oplus W$, we have

$$\langle d(x_1, x_2), (y_1, y_2) \rangle_V = \langle dx, y \rangle_V = \langle x, d^\dagger y \rangle_V = \langle (x_1, x_2), d^\dagger(y_1, y_2) \rangle.$$

By direct computation, we can work out the matrix of $d^\dagger \in M_2(\overline{\mathbb{Q}_\ell})$ should be in terms of that of d , and we find that $\dagger$ is the standard involution on $M_2(\overline{\mathbb{Q}_\ell})$ when Ω_W is symmetric, and the transpose when Ω_W is alternating. The standard involution (which is invariant under base change and change of basis) is a positive involution of D if D is definite, and is not a positive involution if D is indefinite. Since the Rosati involution is a positive involution, the result follows. $\square$

Remark. When A is a “Weil type” type IV abelian fourfold (see e.g. [LPS13, Section 2]), $\text{End}_{\mathbb{Q}}^0(A)$ is an imaginary quadratic field. One can check in this case that $\text{Res}_{G_L} \rho_A \otimes \overline{\mathbb{Q}_\ell} \cong \xi \oplus \xi^\tau$ and that $\xi^\vee \cong \xi^\tau$ – in this case, ξ preserves a unitary pairing, rather than a symplectic or orthogonal one.

All told, each ξ_j in Main Theorem 1 preserves – up to similitude – a symplectic pairing when A is type I or II, an orthogonal pairing when A is type III, and a unitary pairing when A is type IV.

The following corollary addresses a question of [BCGP21b], by showing that their counterexamples to the original version of the paramodular conjecture of Brumer–Kramer (see e.g. [Mar18]) cannot be abelian varieties of type III.

Corollary 3.7. *Let A/F be a geometrically simple abelian fourfold. Suppose that $V_\ell(A) \otimes \overline{\mathbb{Q}_\ell} \cong \rho_\ell \oplus \rho_\ell$ for some strongly irreducible $\rho_\ell: G_{\mathbb{Q}} \rightarrow \text{GSp}_4(\overline{\mathbb{Q}_\ell})$. Then, A is of type II(1).*

Proof. Because A is geometrically simple and $\text{End}_{G_{\mathbb{Q}}} V_{\ell}(A) \otimes \overline{\mathbb{Q}_{\ell}} \cong M_2(\overline{\mathbb{Q}_{\ell}})$, $\text{End}^0(A)$ is a quaternion algebra $D/\mathbb{Q}$. Suppose that D is definite. Let $\mathfrak{hg}_{\ell}$ denote the trace 0 part of $\text{Lie } \rho_{\ell}$. By the Mumford–Tate conjecture, Lemma 3.6 implies that the representation of $\mathfrak{hg}_{\ell}$ on V_{ℓ} preserves an orthogonal pairing. Because ρ_{ℓ} preserves a symplectic pairing up to similitude, the representation of $\mathfrak{hg}_{\ell}$ on V_{ℓ} also preserves an symplectic pairing. Because ρ_{ℓ} is strongly irreducible, the representation of $\mathfrak{hg}_{\ell}$, and so by Schur’s lemma there is a unique nonzero $\mathfrak{hg}_{\ell}$ -invariant pairing on V_{ℓ} up to scaling. This pairing cannot be both orthogonal and symplectic, so we have a contradiction. It follows that D is type II(1). $\square$

4 Geometrically type II abelian varieties

In this section, we discuss some phenomena specific to the case of geometrically type II abelian varieties, where D/E is an indefinite quaternion algebra.

Lemma 4.1. *Let A/F an abelian variety of geometric type II. The $\{\varrho_{\iota}\}$ appearing in Equation (10) (and in particular the $\{\varrho_{\iota_0}\}$ appearing in Corollary 1.3) preserve orthogonal pairings up to similitude and have either cyclic or dihedral projective image.*

Proof. Let E be the center of the quaternion algebra $D := \text{End}_{\overline{F}}^0(A)$. By Lemma 3.6, the $\{\xi_{\iota}\}_{\iota}$ appearing in Equation (10) all preserve symplectic pairings up to similitude. By Lemma 3.5, the $\{\varrho_{\iota}\}_{\iota}$ must hence preserve orthogonal pairings up to similitude. Because the $\{\varrho_{\iota}\}_{\iota}$ are two-dimensional orthogonal representations, they have cyclic or dihedral projective images. $\square$

We recall the notion of a distinguished quadratic subfield of D from [LSSV24, Subsection 3.4]. By Skolem-Noether, the Rosati involution on $d \in D$ has the form $d \mapsto \mu \bar{d} \mu^{-1}$ for some nontrivial $\mu \in D^{\times}/E^{\times}$, where $d \mapsto \bar{d}$ denotes the standard involution. Because D is indefinite, the standard involution on D is not a positive involution, as discussed in the proof of Lemma 3.6, so we must have $\bar{\mu} = -\mu$. Because the Rosati involution is an involution and the center of D is E , $E' := E(\mu)$ is a (well-defined) quadratic extension of E (depending only on A and its polarization). Under this map, $\mu \mapsto \bar{\mu}$. Since $\mu \notin E$, it follows that E'/E is a CM extension.

Lemma 4.2. *Suppose that A/F is such that $\text{End}_{\overline{F}}(A)$ contains an indefinite quaternion algebra D/E for some totally real field E . Then, there exists a quadratic extension F'/F such that $\text{End}_{F'}(A)$ contains the distinguished CM subfield $E' := E(\mu)$ of D .*

Proof. As usual, write ρ_A for the representation on $V_{\ell}(A)$, and let L/F be the minimal extension over which $\text{End}_L^0(A) = \text{End}_{\overline{F}}^0(A)$. Recall that G_L is a normal subgroup of G_F . The Galois action of G_F on D can be understood via the action of the image of D in $\text{End}(V_{\ell}(A))$. An element $d \in D \hookrightarrow \text{End}_{G_L}(V_{\ell}(A))$ is in $\text{End}_{F'}^0(A)$ if and only if it commutes with the image of $G_{F'}$, i.e. if $\rho_A(g') d \rho_A(g')^{-1} = d$ for any $g' \in G_{F'}$. For any $g \in G_F$ and $d \in D$, we have $\rho_A(g) d \rho_A(g)^{-1} \in \text{End}_K^0(A)$ because for any $h \in G_K$,

$$\rho_A(h) \rho_A(g) d \rho_A(g)^{-1} \rho_A(h)^{-1} = \rho_A(g) \rho_A(h') d \rho_A(h')^{-1} \rho_A(g)^{-1} = \rho_A(g) d \rho_A(g)^{-1}.$$

The Rosati involution (or rather, its $\overline{\mathbb{Q}_{\ell}}$ -linear extension to $D \otimes \overline{\mathbb{Q}_{\ell}}$) commutes with the G_F -action on D – for any $v, v' \in V_{\ell}(A)$, letting $\psi(\cdot, \cdot)$ denote the Weil pairing on $V_{\ell}(A)$, we have

$$\psi(\rho_A(g) d \rho_A(g)^{-1} v, v') = \epsilon(g) \psi(d \rho_A(g)^{-1} v, \rho_A(g)^{-1} v') = \epsilon(g) \psi(\rho_A(g)^{-1} v, d^{\dagger} \rho_A(g)^{-1} v') = \psi(v, \rho_A(g) d^{\dagger} \rho_A(g)^{-1} v').$$

However, we also know that $d^\dagger = \mu \bar{d} \mu^{-1}$ for some $\mu \in D \hookrightarrow M_{2g}(\overline{\mathbb{Q}_\ell})$. The standard involution on $M_{2g}(\overline{\mathbb{Q}_\ell})$ is the adjugate action $M \mapsto \det(M)M^{-1}$. As such,

$$(\rho_A(g)d\rho_A(g)^{-1})^\dagger = \overline{\mu\rho_A(g)d\rho_A(g)^{-1}}\mu^{-1} = \mu\rho_A(g)\bar{d}\rho_A(g)^{-1}\mu^{-1}.$$

This is equal to $\rho_A(g)d^\dagger\rho_A(g)^{-1}$ for all $d \in D$ if and only if $\rho_A(g)^{-1}\mu\rho_A(g) = \omega(g)\mu$ for some character $\omega(g) \in E$. Because E is totally real, ω must be quadratic. As such, taking E'/E to be the kernel of ω , $\rho_A(g)^{-1}\mu\rho_A(g) = \mu$ for all $g \in G_{F'}$ and hence $\mu \in \text{End}_{F'}^0(A)$. $\square$

Lemma 4.2 is not true without allowing an extension F'/F – consider for instance GL_2 -type abelian surfaces over $\mathbb{Q}$, which have endomorphisms by a real quadratic field over $\mathbb{Q}$ but acquire quaternionic multiplication over a quadratic extension.

lem:im-real Lemma 4.3. *Suppose that $\text{Res}_{G_{F'}} V_\ell(A) \otimes \overline{\mathbb{Q}_\ell} \cong \bigoplus_{\iota: E' \rightarrow \overline{\mathbb{Q}_\ell}} \xi'_\iota \otimes (1 \oplus \eta_\iota)$, where $E' := \text{End}_{F'}^0(A)$ and the ξ'_ι are strongly irreducible and self-dual up to similitude. Then, the pairing induced by the Weil pairing on $(1 \oplus \eta_\iota)$ (given by Lemma 3.5) is*

1. $e_1 \otimes e_1 + e_2 \otimes e_2$ if E'/E is totally real;
2. $e_1 \otimes e_2 + e_2 \otimes e_1$ if E'/E is CM,

where e_1 and e_2 are basis vectors for the subrepresentations 1 and η_ι , respectively.

For example, in the context of Corollary 1.3, Lemma 4.3 applies to abelian varieties A/F with fields F'/F such that $\text{End}_{F'}^0(A) = E' \supseteq E$, in which case $\text{Res}_{G_{F'}} \rho_\iota \cong 1 \oplus \eta_\iota$ with $\eta_\iota \neq 1$.

Proof. The subspaces ξ_ι and $\xi_\iota \otimes \eta$ are the eigenspaces of the E' -action on $V_\ell(A)$ with respect to the two embeddings $\tilde{\iota}, \tilde{\iota}': E' \hookrightarrow \overline{\mathbb{Q}_\ell}$ extending ι . By Lemma 3.4, the Weil pairing restricts to a pairing on $\xi_\iota \otimes (1 \oplus \eta_\iota)$. The Rosati involution $\dagger: E' \rightarrow E'$ – which is trivial if E' is totally real and is complex conjugation if E' is CM – is the adjoint operation for $e \in E'$ with respect to the Weil pairing. Writing $\psi(\cdot, \cdot)$ for the Weil pairing this means that for any $e \in E'$, $x \in \xi_\iota$, and $y \in \xi_\iota \otimes \eta$,

$$\tilde{\iota}(e)\psi(x, y) = \psi(ex, y) = \psi(x, e^\dagger y) = \tilde{\iota}'(e^\dagger)\psi(x, y).$$

In particular, if E' is totally real, $\dagger$ is trivial, and $\psi(x, y) = 0$. Similarly, taking $x, y \in \xi_\iota$,

$$\tilde{\iota}(e)\psi(x, y) = \psi(ex, y) = \psi(x, e^\dagger y) = \tilde{\iota}(e^\dagger)\psi(x, y).$$

In particular, if E is imaginary quadratic, $\dagger$ is nontrivial, and $\psi(x, y) = 0$. We can do the same computation for $x, y \in \xi_\iota \otimes \eta$. $\square$

Known automorphy theorems that we might want to apply to the compatible system $\mathcal{R}$ over a totally real field typically impose a parity condition on the representations in $\mathcal{R}$. For example, the modularity lifting theorem of Boxer–Calegari–Gee–Pilloni (Theorem 2.5) for four-dimensional symplectic Galois representations, which we wish to apply to prove potential modularity of type I(2) and II(1) abelian fourfolds, requires $\mathcal{R}$ to be odd – in this case, the associated automorphic forms are Siegel modular forms. However, there are in fact abelian varieties whose associated compatible systems $\mathcal{R}$ are even (Section 5.4). As such, we will content ourself with the following weaker result.

lem:odd Lemma 4.4. *Suppose that A/F is geometrically type II. Assume that the field K in Main Theorem 1 be totally real, and there that is no totally real K'/K for which $\text{End}_{K'}(A)$ is a CM extension of E . Then, the $\{\xi_\iota\}$ appearing in Equation (10) (and hence, the compatible system $\mathcal{R}$ appearing in Main Theorem 1) are (is) odd.*

Note that because the $\{\xi_\ell\}_\ell$ are representations of G_K , the question of oddness only makes sense when K is totally real. The third condition of Lemma 4.4 may seem unnatural, but Main Theorem 3 demonstrates that it is in fact tight.

Proof. Fix an embedding $\iota: E \hookrightarrow \overline{\mathbb{Q}_\ell}$, and let $\xi := \xi_\ell$ and $\varrho := \varrho_\ell$. As discussed earlier, ϱ preserves an orthogonal pairing (up to similitude). The similitude character of ϱ is then a one-dimensional summand in $\text{Sym}^2 \varrho$.

First suppose that ϱ has cyclic projective image, so – up to possibly replacing ξ by a twist – $\varrho \cong 1 \oplus \eta$. Then,

$$\text{Sym}^2 \varrho \cong 1 \oplus \eta^2 \oplus \eta. \quad (12) \quad \text{\texttt{\textcolor{blue}{\code{eq:symp-odd-2}}}}$$

The first two direct summands in Equation (12) are automatically even. The similitude character can only be η when the direct summands of ϱ corresponding to 1 and η are paired against each other by the Weil pairing, which by Lemma 4.3 can happen only if $\text{End}_K^0(A)$ is a CM extension of E . Under our assumptions, any possible similitude character in Equation (12) is thus even.

Suppose instead that ϱ has dihedral projective image, so $\varrho \cong \text{Ind}_{G_{K'}}^{G_K} \eta$ for K'/K a quadratic extension. Let M/K' be the field cut out by η . Considering the restriction to $G_{K'}$ and the determinant, one can show that

$$\text{Sym}^2 \text{Ind}_{G_{K'}}^{G_K} \eta \cong \text{Ind}_{G_{K'}}^{G_K} \eta^2 \oplus (\eta \circ \text{Ver}), \quad (13) \quad \text{\texttt{\textcolor{blue}{\code{eq:symp-odd-3}}}}$$

where $\text{Ver}: G_K^{\text{ab}} \rightarrow G_{K'}^{\text{ab}}$ is the transfer map. Here, $\eta \circ \text{Ver}$ is the similitude character of a pairing that is given (in an appropriate basis) by the matrix $\begin{pmatrix} & 1 \\ 1 & \end{pmatrix}$. If $\eta^2 \not\cong \eta_\sigma^2$ and $\text{Ind}_{G_{K'}}^{G_K} \eta^2$ is irreducible, the similitude character of the orthogonal pairing on ϱ is necessarily $\eta \circ \text{Ver}$ as it is the only one-dimensional summand in Equation (13). We have

$$(\eta \circ \text{Ver})(c) = \begin{cases} \eta(c)\eta(\sigma c \sigma^{-1}) & c \in G_{K'} \\ \eta(c^2) & c \notin G_{K'}. \end{cases}$$

In the second case, $\eta(c^2) = \eta(1) = 1$. In the first case, K' is totally real and $\text{Res}_{G_{K'}} V_\ell(A)$ has cyclic projective image, so ξ is odd by our assumptions and our discussion of the cyclic projective image case.

If $\eta^2 \cong \eta_\sigma^2$, then the projective image of ϱ is D_2 , and additionally η^2 extends to a character $\tilde{\eta}^2$ of G_K . In this case, ϱ can be induced from three different index 2 subgroups of G_K , corresponding to the three index 2 subgroups of D_2 . By Lemma 4.2, one of the corresponding extensions K'/K is such that $\text{End}_{K'}^0(A)$ is an imaginary quadratic field. We have

$$\text{Sym}^2 \text{Ind}_{G_{K'}}^{G_K} \eta \cong \tilde{\eta}^2 \oplus \left(\tilde{\eta}^2 \otimes \omega_{K'/K} \right) \oplus (\eta \circ \text{Ver}). \quad (14) \quad \text{\texttt{\textcolor{blue}{\code{eq:symp-odd-1}}}}$$

Because $\text{End}_{K'}^0(A)$ is an imaginary quadratic field, the similitude character of the orthogonal pairing must be $\eta \circ \text{Ver}$ by Lemma 4.3. The result then follows from the earlier discussion. $\square$

4.1 Fields of definition of endomorphisms of abelian varieties

Main Theorem 1 lets us reinterpret some facts about the field of definition of endomorphisms of abelian varieties in the language of Galois representations.

The compatible systems $\mathcal{R}$ and $\mathcal{A}$ appearing in Main Theorem 1 are only canonical up to twist. Nonetheless, it makes sense to talk about the projective kernel of $\mathcal{A}$, or equivalently, the shared

projective kernel of the $\{\text{Proj } \varrho_{\iota_0}\}_{\iota_0}$ appearing in Corollary 1.3. The field fixed by this projective kernel is exactly the field over which $\text{End } V_\ell(A) \otimes \overline{\mathbb{Q}}_\ell \cong M_r(\overline{\mathbb{Q}}_\ell)^e$, i.e. the field over which the geometric endomorphisms of A are defined.

As in Main Theorem 1, let K the minimal field of definition of the endomorphisms in E . By Skolem-Noether, $\text{Gal}(L/K)$ embeds into $D^\times/E^\times$. Picking an embedding of E into $\mathbb{C}$ gives us an embedding of $D^\times/E^\times$ into $\text{PGL}_2(\mathbb{C})$. When A is type II, choosing an embedding of E into $\mathbb{R}$ induces an embedding of $D^\times/E^\times \hookrightarrow \text{PGL}_2(\mathbb{R})$, and the only finite subgroup of $\text{PGL}_2(\mathbb{R})$ are cyclic or dihedral. Therefore, any finite subgroup of $D^\times/E^\times$ is necessarily cyclic or dihedral when D is indefinite, and hence $\text{Gal}(L/K)$ is cyclic or dihedral for geometrically type II abelian varieties (this is part of a much stronger result proved in [CF00, Lemma 2.8]). Lemma 4.1 gives an alternative proof of this fact – it reflects the Artin compatible system $\mathcal{A}$ being of orthogonal type which in turns follows from $\mathcal{R}$ being of symplectic type.

Remark. There are abelian varieties where $E \subsetneq \text{End}_K^0(A)$. Let $f \in S_2(\Gamma_0(243))$ be the newform (243.2.a.d)

$$f := q + \beta q^2 + 4q^4 - \beta q^5 + 2q^7 + 2\beta q^8 - 6q^{10} + \beta q^{11} - q^{13} + 2\beta q^{14} + \cdots,$$

which has coefficients over $\mathbb{Q}(\sqrt{6})$. Let A be the abelian variety of GL_2 -type associated to f , so $\rho_A \otimes \overline{\mathbb{Q}}_\ell \cong \rho_{f,\ell} \oplus \rho_{f^\tau,\ell}$. This f admits the inner twist $f^\tau = f \otimes \left(\frac{-3}{\cdot}\right)$, so writing $L := \mathbb{Q}(\sqrt{-3})$, $\text{Res}_L V_\ell(A) \otimes \overline{\mathbb{Q}}_\ell \cong (\text{Res}_L \rho_f)^{\oplus 2}$. In particular, $\text{End}_{\mathbb{Q}}^0(A) \otimes \overline{\mathbb{Q}} \cong M_2(\overline{\mathbb{Q}})$, so $\text{End}_L(A)$ is either $M_2(\mathbb{Q})$ or a quaternion algebra D with center $\mathbb{Q}$. Looking at the Fourier coefficients of f , we see that $a_p(f) \in \mathbb{Q}$ whenever p is split in L , so $\text{Res}_L \rho_f$ has traces in $\mathbb{Q}$. If it were defined over $\mathbb{Q}_\ell$ for all ℓ , then $\text{Res}_L \rho_f$ would be the Galois representation associated to an elliptic curve E/F that is isogenous to its Galois conjugate (a so-called $\mathbb{Q}$ -curve), A would be geometrically reducible, and $\text{End}_F^0(A)$ would be $M_2(\mathbb{Q})$. However, it turns out that ρ_f is not defined over $\mathbb{Q}_\ell$ for all ℓ , and $\text{End}_L^0(A)$ is the quaternion algebra $D/\mathbb{Q}$ with discriminant 6. In particular, $\text{End}_{\mathbb{Q}}^0(A) \cong \mathbb{Q}(\sqrt{6})$, $E := \mathbb{Q}$, and D is the aforementioned quaternion algebra.

5 Applications to abelian fourfolds

In this section, we apply Main Theorem 1 to prove Main Theorem 2. By Corollary 3.3, we immediately deduce potential modularity of abelian fourfolds of geometric types I(4), II(2), IV(1,2), and IV(1,4). It remains to address geometric types I(2) and II(1). For ease of exposition, we first give the proof in the case of geometrically type II(1) fourfolds, and afterwards describe the modifications needed for type I(2) fourfolds.

5.1 Potential modularity of type II(1) abelian fourfolds

Let A/F be a type II(1) abelian fourfold with $D := \text{End}_F^0(A)$, and let $\mathcal{V} := \mathcal{V}(A) = (F; \mathbb{Q}, S, (Q_{\mathcal{V},v}(X))_v, (\rho_{A,\ell})_\ell)$ be the compatible system of 8-dimensional representations associated to A . By Main Theorem 1, $E_0 = E = \mathbb{Q}$ and $\mathcal{V} \cong \text{Des}_{\mathbb{Q}}(\mathcal{R} \otimes \mathcal{A})$, where $\mathcal{R} = (F; E', S, (Q_{\mathcal{R},v}(X))_v, (r_\lambda)_\lambda)$ is a compatible system of four-dimensional symplectic representations and $\mathcal{A} = (F; E', S, (Q_{\mathcal{A},v}(X))_v, (a_\lambda)_\lambda)$ is a compatible system of two-dimensional Artin representations, both (after possibly enlarging coefficients) with coefficients in some $E'/\mathbb{Q}$. Let S_0 be the set of rational primes divisible by primes in S . For each rational prime ℓ , we pick a prime $\lambda|\ell$ in E' and define $\xi_\ell := r_\lambda$ and $\varrho_\ell := a_\lambda$, so that $\rho_{A,\ell} \otimes \overline{\mathbb{Q}}_\ell \cong \xi_\ell \otimes \varrho_\ell$ for all $\ell \notin S_0$ – because $\mathcal{R} \otimes \mathcal{A}$ descends to $\mathbb{Q}$, the specific choice does not matter. As before, we write M/F for the extension through which $\mathcal{A}$ factors. In this subsection, we prove the following theorem.

Theorem 1.4. *Let F be a totally real field and A/F be an odd type abelian fourfold with $\text{End}_{\mathbb{Q}}^0(A) = D$, where D is an indefinite quaternion algebra with center $\mathbb{Q}$. Then, there exists a compatible system $\mathcal{R}$ of four-dimensional strongly irreducible symplectic (odd) representations and a compatible system $\mathcal{A}$ of two-dimensional Artin representations with cyclic or dihedral projective image such that*

1. $\mathcal{R} \otimes \mathcal{A}$ descends to $\mathbb{Q}$;
2. $\mathcal{V}(A) \cong \text{Des}_{\mathbb{Q}}(\mathcal{R} \otimes \mathcal{A})$; and
3. A is potentially modular, and is modular whenever any r_{λ} in $\mathcal{R}$ is modular.

By Main Theorem 1 and Corollary 3.2, it suffices to prove that $\mathcal{R}$ is potentially automorphic.

Definition 5.1 (Following [BCGP21b, Definition 9.2.3]). We say that a prime v of F is good for A/F if

- The image of $\bar{\xi}_p$ contains $\text{Sp}_4(\mathbb{F}_p)$;
- $(\xi_p)|_{G_{F_v}}$ is p -distinguished weight 2 ordinary, as defined in [BCGP21b, Definition 7.3.1];
- $p \nmid [M : F]$;
- p splits completely in the extension M/F , where M is the extension through which the Artin compatible system $\mathcal{A}$ factors; and
- p is split in D .

We say that a rational prime p is good for A/F if p splits completely in $F/\mathbb{Q}$ and if, for all primes $v|p$ in F , v is good.

Lemma 5.2. *Let A be a geometrically type II(1) abelian fourfold. The set of good rational primes for A/F has relative density 1 in the set of rational primes which split completely in $M/\mathbb{Q}$, and in particular, have positive relative density in the set of all rational primes.*

Proof. It suffices to show that a density 1 subset of the primes of M are good. By Main Theorem 1, $\text{Res}_M \mathcal{V} \cong \text{Res}_M \mathcal{R}^{\oplus 2}$. Let $A' := A_M$, $\mathcal{V}' := \text{Res}_M \mathcal{V}$ and $\mathcal{R}' := \text{Res}_M \mathcal{R}$. Let $\mathcal{R}' = (r'_\ell)_\ell$. For any finite place v of M , $Q_{\mathcal{V}',v}(X) = Q_{\mathcal{R}',v}^2$. In particular, because $Q_{\mathcal{V}',v}(X) \in \mathbb{Z}[X]$ and both $Q_{\mathcal{V}',v}(X)$ and $Q_{\mathcal{R}',v}(X)$ are monic, $Q_{\mathcal{R}',v}(X) \in \mathbb{Z}[X]$ for such v .

The remainder of the proof is largely the same as the proof of [BCGP21b, Lemma 9.2.5], but because this is the lemma which requires modification in the case of geometrically type I(2) fourfolds later, we spell out some of the details. Because $\rho_{A'} \cong \rho_{A'}^\vee \otimes \epsilon$, the roots of $Q_{\mathcal{R}',v}(X)$ have the form $\{\alpha_v, \beta_v, p\alpha_v^{-1}, p\beta_v^{-1}\}$, and we can write $Q_{\mathcal{R}',v}(X) := x^4 - a_1(\text{Frob}_v)x^3 + a_2(\text{Frob}_v)x^2 - pa_1(\text{Frob}_v)x + p^2$. We representation $(r'_p)|_{G_{F_v}}$ is ordinary (at v) if and only if $v_p(a_2) = 0$ (equivalent to $v_p(\alpha_v) = v_p(\beta_v) = 0$). Further, the representation $(r'_p)|_{G_{F_v}}$ is p -distinguished ordinary if and only if $v_p(a_1(\text{Frob}_v)^2 - 4a_2(\text{Frob}_v)) = 0$ (equivalent to $v_p(\alpha_v - \beta_v) > 0$).

These criteria let us translate p -distinguished ordinarity – a condition on $r'_p(\text{Frob}_v)$ – into a condition on $Q_{\mathcal{R}',v}(X)$, independent of the representation r'_p . By the Weil bounds, for any embedding $j: \overline{\mathbb{Q}} \hookrightarrow \mathbb{C}$, $|j(\alpha_v)| = |j(\beta_v)| = \sqrt{p}$. Therefore, $|a_1(\text{Frob}_v)| \leq 4\sqrt{p}$ and $|a_2(\text{Frob}_v) - 2p| \leq 4p$. In particular, v fails to be ordinary only if $a_2(\text{Frob}_v) = cp$ for some $c \in \mathbb{Z} \cap [-6, 6]$, and fails to be p -distinguished ordinary if and only if $a_1(\text{Frob}_v)^2 = 4a_2(\text{Frob}_v) + c'p$ for some $c' \in \mathbb{Z} \cap [-40, 40]$. Let G_ℓ be the ℓ -adic algebraic monodromy group of A' – the Zariski closure (over $\mathbb{Q}_\ell$) of the image of $\rho_{A'}$ in $\text{GSp}_{2g}(\mathbb{Q}_\ell)$. The Mumford–Tate group G of a type II abelian fourfold is a twisted form G of $\text{GSp}_{4,\mathbb{Q}}$.

which splits over $\mathbb{Q}_\ell$ for any ℓ unramified in D [?]. Fix ℓ to be such a prime. The Mumford–Tate conjecture is known for type II abelian fourfolds [MZ95], so $G_\ell = G \times_{\mathbb{Q}} \mathbb{Q}_\ell = \mathrm{GSp}_{4, \mathbb{Q}_\ell}$. In particular, we deduce that $r'_\ell: G_F \rightarrow G_\ell$ has Zariski dense image in $\mathrm{GSp}_{4, \mathbb{Q}_\ell}$. Consider $\rho := \rho_\ell := (r'^{\otimes 2} \oplus \wedge^2 r') \otimes \varepsilon^{-1}$. This representation factors through $\rho: G_M \rightarrow G_\ell \hookrightarrow \mathrm{GSp}_{22}(\mathbb{Q}_\ell)$. For $g \in \mathrm{GSp}_{22}(\mathbb{Q}_\ell)$, let $\mathrm{tr}_i(g)$, $i \in \{1, 2\}$, be the trace of the i^{th} block in ρ_ℓ . Because $a_1^2(\mathrm{Frob}_v) = \mathrm{tr} r'^{\otimes 2}$ and $a_2(\mathrm{Frob}_v) = \mathrm{tr} \wedge^2 r'$, v fails to be ordinary if and only if $\rho(\mathrm{Frob}_v)$ lands in the union Y of the hypersurfaces cut out by the equations

$$\{\mathrm{tr}_2(g) = c: c \in \mathbb{Z} \cap [-6, 6]\} \quad \text{and} \quad \{\mathrm{tr}_1(g) - 4\mathrm{tr}_2(g) = c': c' \in \mathbb{Z} \cap [-40, 40]\}. \quad (15) \quad \{\text{eq:good-1}\}$$

Because Y does not contain $\mathrm{im} G_\ell$, the preimage Z of Y under $G_\ell \hookrightarrow \mathrm{GSp}_{22}(\mathbb{Q}_\ell)$ is a subvariety of positive codimension. The Chebotarev density theorem applied to $\rho: G_F \rightarrow G_\ell$ then implies that Frob_v avoids Z (and hence that v is ordinary and distinguished p -ordinary) for a density 1 subset of primes of M . Lastly, because $\mathcal{R}'$ satisfies the Mumford–Tate conjecture, [?, Theorem 3.17] implies that $\overline{r'_p}$ has image containing $\mathrm{Sp}_4(\overline{\mathbb{F}}_p)$ for all sufficiently large primes p . $\square$

Theorem 5.3. *Let A/F be an abelian variety over a totally real field and p and q good primes for A in the sense of Definition 5.1. There exist a totally real field extension $\tilde{F}/F$ and a quadratic extension F_1/F such that*

- p and q split completely in $\tilde{F}/\mathbb{Q}$; and
- $\tilde{F}/F$ and F_1/F are linearly disjoint from each other, from L/F , and from the kernels of the actions of G_F on $A[p]$ and $A[q]$.

and a Galois representation $\overline{\rho}_q^{\mathrm{m}}: G_{\tilde{F}} \rightarrow \mathrm{GSp}_4(\mathbb{F}_q)$ with cyclotomic similitude such that

- $\overline{\rho}_q^{\mathrm{m}}$ is automorphic;
- For any complex conjugation in $G_{\tilde{F}}$, $\overline{\rho}_q^{\mathrm{m}}(c) \sim \mathrm{diag}(1, 1, -1, -1)$;
- $\left(\overline{\rho}_q^{\mathrm{m}}|_{G_{\tilde{F}_v}}\right)^{\oplus 2} \cong (\xi_q|_{G_{F_v}})^{\oplus 2} \cong A[q]^\vee|_{G_{\tilde{F}_v}}$ for all places $v|p$;
- $\left(\overline{\rho}_q^{\mathrm{m}}|_{G_{\tilde{F}_w}}\right)^{\oplus 2} \cong \left((\xi_q|_{G_{\tilde{F}_w}})^{\mathrm{ss}}\right)^{\oplus 2} \cong \left(A[q]^\vee|_{G_{\tilde{F}_w}}\right)^{\mathrm{ss}}$ for all places $w|q$; and
- $\overline{\rho}_q^{\mathrm{m}}$ is vast and tidy in the sense of [BCGP21b, Definitions 7.5.6 and 7.5.11]

Proof. This follows from the proof of [BCGP21b, Theorem 9.2.7] with their $\overline{\rho}_{A,q}$ (for them A is an abelian surface) replaced with our $\overline{\xi}_q$. Note that because q is good, $(\xi_q)|_{G_{F_w}}$ is q -distinguished weight 2 ordinary. $\square$

Lemma 5.4. *Let q be a rational prime split completely in F and not dividing the conductor of A . Let w be a prime of F above q . There exists an abelian variety A^{can}/F_w such that for all primes w' of L above w ,*

1. $\mathrm{End}_{F_w}^0(A^{\mathrm{can}}) = \mathrm{End}_{F_w}^0(A)$ as $\mathrm{Gal}(L_{w'}/F_w)$ -modules.
2. $A^{\mathrm{can}}[p]^\vee \cong (A[p]|_{G_{F_w}})^\vee$
3. $A^{\mathrm{can}}[q]^\vee \cong (A[q]|_{G_{F_w}})^{\mathrm{ss}}$.

Note that because L/F is Galois, $L_{w'}/F_w$ is Galois and independent of w' .

Proof. We take A^{can}/F_w to be the Serre–Tate canonical lift of A_{F_w} . As in the proof of Lemma 5.2, let $\{\alpha_w, \beta_w, p\alpha_w^{-1}, p\beta_w^{-1}\}$ be the eigenvalues (each with multiplicity 2) of $\rho_{A,q}(\text{Frob}_w)$. The defining property of the Serre–Tate canonical lift (see e.g. [?]) is that

$$\rho_{A^{\text{can}},q} \cong \rho_{A,q}^{\text{ss}} \sim \begin{pmatrix} \lambda_{\bar{\alpha}_w} & & & \\ & \lambda_{\bar{\beta}_w} & & \\ & & \varepsilon \lambda_{\bar{\alpha}_w}^{-1} & \\ & & & \varepsilon \lambda_{\bar{\beta}_w}^{-1} \end{pmatrix}^{\oplus 2}.$$

The third property follows immediately.

Because A^{can}/F_w has good reduction and residue field characteristic q , its ℓ -adic Galois representation is unramified for any $\ell \neq q$, and the eigenvalues of Frob_w do not depend on ℓ . Furthermore, they are the same as the Frobenius eigenvalues of A_{F_w} . Taking $\ell = p$, we therefore have

$$A^{\text{can}}[p] \sim \begin{pmatrix} \lambda_{\bar{\alpha}_w} & & & \\ & \lambda_{\bar{\beta}_w} & & \\ & & \varepsilon \lambda_{\bar{\alpha}_w}^{-1} & \\ & & & \varepsilon \lambda_{\bar{\beta}_w}^{-1} \end{pmatrix}^{\oplus 2} \sim A_{F_w}[p].$$

The first property follows from [Mes72, Corollary 3.4]. $\square$

We are now ready to prove the third part of Theorem 1.4. We follow the approach (and where possible, the notation) of [BCGP21b, Section 9.2]. Let p and q be rational primes which are good for A/F in the sense of Definition 5.1 – such p and q exist by Lemma 5.2. Because p and q are split in M/F , for any primes $v|p$ and $w|q$ in F , the Artin factor in $\mathcal{V} \cong \mathcal{R} \otimes \mathcal{A}$ is trivial. Hence, for any prime ℓ , $(\rho_{A,\ell})|_{G_{F_v}} \cong ((\xi_v, \ell))^{\oplus 2}$ (and similarly for $w|q$), where $\xi_{v,\ell}: G_{F_v} \rightarrow \text{GSp}_4(\overline{\mathbb{Q}}_\ell)$ has big image, Hodge–Tate weights $(0, 0, 1, 1)$, and where $\xi|_{G_{F_\lambda}}$ is de Rham for any $\lambda|\ell$.

For ease of notation, for the remainder of this proof we replace F with the extension $\tilde{F}/F$ appearing in Theorem 5.3, A with its base change to $\tilde{F}$, L with $L\tilde{F}$, and the extension F_1/F in Theorem 5.3 with $F_1\tilde{F}/\tilde{F}$. Note that $\text{Gal}(L\tilde{F}/\tilde{F}) = \text{Gal}(L/F)$.

The main difference between this proof and the approach of [BCGP21b, Section 9.2] is that we replace the moduli of abelian surfaces with a twisted moduli space of type II abelian fourfolds capturing the Galois action on the endomorphism algebra.

Let Y'_A/F be the scheme given by applying Theorem 2.14 with $F' = F$, $L' = L$, $\rho_{\text{End}}: \text{Gal}(L/F) \rightarrow \text{Aut}(\text{End}_L^0(A)) \cong \text{Aut}(D)$ the map given by the Galois action on the geometric endomorphisms of A , $\bar{\rho}_p := A[p]^\vee$, and $\bar{\rho}_q := \bar{\rho}_q^{\text{mc}}$ (as defined in Theorem 5.3), and allowing enough auxiliary level structure to ensure that the corresponding untwisted PEL Shimura variety Y is smooth. Because the class number of $E = \mathbb{Q}$ is 1, the untwisted PEL Shimura variety is also geometrically connected, so by Theorem 2.14, Y'_A is also smooth and geometrically connected.

We want to apply the Moret–Baily theorem (see e.g. [BLGGT14, Proposition 3.1.1]) to Y'_A to find a totally real field F''/F in which p and q split completely and such that Y'_A has a point over F'' with good ordinary reduction at p and q . Additionally, we want F''/F to be linearly disjoint from F_1/F , L/F , and the kernels of the G_F actions on $A[p]$ and $A[q]$. To impose the local conditions upon F''/F , we require $\Omega'_{A,u} := Y'_A(F_u)^{\text{ord}}$, $\Omega'_{A,v} := Y'_A(F_v)^{\text{ord}}$, and $\Omega'_{A,w} := Y'_A(F_w)$ to be nonempty for every real place $u|\infty$, every $v|p$, and every $w|q$. We check these in turn.

Because A is defined over a totally real field, $\rho_{\text{End}}(c_u) = \rho_{\text{End},A}(c_u)$ (because $\rho_{\text{End}} = \rho_{\text{End},A}$) $\bar{\rho}_p(c_u) = A[p]^\vee(c_u)$ (because $\bar{\rho}_p = A[p]^\vee$) and $\bar{\rho}_q(c_u) \cong A[q]^\vee(c_u)$ (by the second property of $\bar{\rho}_q^{\text{mc}}$ in Theorem 5.3), A_{F_u} defines a point in $\Omega'_{A,u}$ for every $u|\infty$. Similarly, because A is defined over a

field where p splits completely and $\bar{\rho}_q|_{G_{F_v}} \cong A[q]^\vee|_{G_{F_v}}$ (by the third property of $\bar{\rho}_q^{\text{m}}$ in Theorem 5.3), A_{F_v} defines a point in $\Omega'_{A,v}$ for every $v|p$. Lastly, because A is defined over a field where q splits completely, for any $w|q$ the abelian variety A^{can}/F_w provided by Lemma 5.4 defines a point in $\Omega'_{A,w}$ for every $w|q$.

Therefore, by the Moret-Baily theorem, Y'_A has a point over F'' . Such a point corresponds, by Theorem 2.14, to an abelian fourfold B/F'' with ordinary good reduction at p and q such that such that (along with some information about the endomorphisms of B)

$$B[p]^\vee \cong A[p]^\vee|_{G_{F''}} \quad \text{and} \quad B[q]^\vee \cong \left(\bar{\rho}_q^{\text{m}}|_{G_{F''}}\right)^{\oplus 2}.$$

Let $\mathcal{R}_B$ be the compatible system of Galois representations associated by Main Theorem 1 to B , and let for each rational prime ℓ let $\{\xi_{B,\ell}\}_\ell$ and $\{\varrho_{B,\ell}\}_\ell$ be the ℓ -adic representations given by Corollary 1.3. We have

$$\left(\bar{\rho}_q^{\text{m}}|_{G_{F''}}\right)^{\oplus 2} \cong \overline{\rho_{B,q}} \cong \overline{\xi_{B,q}} \otimes \overline{\varrho_{B,q}}.$$

Because F''/F and F_1/F are linearly disjoint, $\bar{\rho}_q^{\text{m}}$ is irreducible, and so $\overline{\varrho_{B,q}}$ is a character $\overline{\eta_q}$ and $\overline{\xi_{B,q}} \cong \bar{\rho}_q^{\text{m}}|_{G_{F''}} \otimes \overline{\eta_q}^{-1}$. Therefore, $\overline{\xi_{B,q}}$ is modular. Additionally, because $\bar{\rho}_q^{\text{m}}|_{G_{F''}}^{\oplus 2} \cong A[q]^\vee|_{G_{F''}}$, $\xi_{B,q}$ is not only ordinary at w (which we already know from $B \in \Omega'_{A,w}$) but q -distinguished ordinary. With the rest of Theorem 5.3 in hand, Theorem 2.5 implies that $\xi_{B,q}$, and hence $\mathcal{R}_B$, is automorphic. From this, we deduce the automorphy of $\xi_{B,p}$. Let $\xi'_{A,p} := \xi_{A,p}|_{G_{F''}}$. We have

$$\overline{\xi_{B,p}} \otimes \overline{\varrho_{B,p}} \cong B[p]^\vee \cong A[p]^\vee \cong \overline{\xi'_{A,p}} \otimes \overline{\varrho_{A,p}}.$$

We want to use Frobenius reciprocity to argue that $\overline{\xi'_{A,p}}$ is automorphic, but need to be somewhat careful because we are working in characteristic p . Nonetheless,

$$\begin{aligned} \text{Hom}_{\overline{\mathbb{F}}_p[G_M]}(\overline{\xi'_{A,p}}, \overline{\xi_{B,p}}) &\cong \text{Hom}_{\overline{\mathbb{F}}_p[G_F]} \left(\overline{\xi'_{A,p}} \otimes_{\overline{\mathbb{F}}_p[G_M]} \overline{\mathbb{F}}_p[G_F], \overline{\xi_{B,p}} \right) \\ &\cong \text{Hom}_{\overline{\mathbb{F}}_p[G_F]} \left(\overline{\xi'_{A,p}} \otimes_{\overline{\mathbb{F}}_p[G_F]} \overline{\mathbb{F}}_p[\text{Gal}(M/F)], \overline{\xi_{B,p}} \right). \end{aligned}$$

Because p is good for A , $p \nmid [M:F]$, so $\overline{\mathbb{F}}_p[\text{Gal}(M/F)]$ is a direct sum of irreducible representations. Furthermore, $\overline{\xi'_{A,p}}$ has big image and is hence irreducible. Because $\text{Res}_{G_M} \overline{\xi_{B,p}} \cong \text{Res}_{G_M} \overline{\xi'_{A,p}}$, $\xi_{B,p}$ is also irreducible. Therefore, $\overline{\xi'_{A,p}}$ is isomorphic to a twist of $\overline{\xi_{B,p}}$ by a character of $\text{Gal}(M/F)$, and is hence is also automorphic. Again applying Theorem 2.5, we deduce that $\xi'_{A,p}$ is automorphic and hence that $\mathcal{R}$ is potentially automorphic, as desired.

Remark. Gu [Gu25] proves the potential modularity of some four-dimensional compatible systems of symplectic type associated to abelian varieties. In particular, [Gu25, Theorem 6.11] implies the potential modularity of type II abelian fourfolds A/F , F a totally real field, for which $\text{End}_F(A)$ is a quaternion algebra with center $\mathbb{Q}$. In general, the minimal field over which the geometric endomorphisms of A are defined need not be totally real, and in these cases Theorem 1.4 does not follow from his theorem. However, his methods, like ours, follow the approach of [BCGP21b, Section 9.2]. Because he is interested in abelian varieties with quaternionic multiplication defined over a general totally real field, he is able to work with real points on the moduli of such abelian varieties directly without worrying about the issue of Galois actions on the endomorphism algebra.

5.2 Potential modularity of geometrically type I(2) abelian fourfolds

subsec:I-2-mod

Let A be a geometrically type I(2) abelian fourfolds. By Main Theorem 1, there exists a compatible system $\mathcal{R}$ of four-dimensional strongly irreducible symplectic representations with coefficients in E such that either

$$\mathcal{V}(A) \cong \text{CRes}_{E/\mathbb{Q}} \mathcal{R} \quad \text{or} \quad \mathcal{V}(A) \cong \text{Des}_{\mathbb{Q}}(\text{Ind}_K^F \mathcal{R}). \quad (16) \quad \{\text{eq:I-2}\}$$

A key distinction from the type II(1) case is that, by Lemma 3.4, $\mathcal{R}$ has cyclotomic similitude, and is in particular odd. Suppose that F is totally real. We call A/F challenging if it is in the second case of Equation (16) and K/F is not totally real.

Theorem 5.5. *Let F be a totally real field and A/F be an abelian fourfold with $\text{End}_{\mathbb{Q}}^0(A) = E$, where E is a real quadratic field. Then, there exists a compatible system $\mathcal{R}$ of four-dimensional strongly irreducible symplectic (odd) representations with coefficients in E such that*

1. $\mathcal{V}(A) \cong \text{CRes}_{E/\mathbb{Q}} \mathcal{R}$ or $\mathcal{V}(A) \cong \text{Des}_{\mathbb{Q}}(\text{Ind}_K^F \mathcal{R})$;
2. A is potentially modular unless we are in the second case above and K/F is not totally real; and
3. A is modular whenever any r_{λ} in $\mathcal{R}$ is modular.

The proof of Theorem 5.5 is largely the same as the approach used in Section 5.1 to prove Theorem 1.4, but with a few deviations.

- Because the compatible system $\mathcal{R}$ has coefficients in a real quadratic field E rather than in $\mathbb{Q}$, we need a slightly different argument (Lemma 5.6) to prove that there are enough good primes.
- Because there is no Artin compatible system $\mathcal{A}$, $\mathcal{R}$ is guaranteed to be odd, so we do not need an “odd type” condition in Theorem 5.5.
- If A is challenging, then because $\mathcal{R}$ is not over a totally real field, we cannot apply Theorem 2.5. When A is not challenging, then by replacing A with A_K we may assume that $\mathcal{V}(A) \cong \text{CRes}_{E/\mathbb{Q}} \mathcal{R}$ for the purposes of proving a potential modularity theorem. In particular, we can assume that A is type I(2) (and not just geometrically type I(2)).
- For a type I(2) abelian fourfold, we have $\rho_{A,\ell} \otimes \overline{\mathbb{Q}_{\ell}} \cong \xi_{1,\ell} \oplus \xi_{2,\ell}$, where the Frobenius characteristic polynomials of $\xi_{1,\ell}$ and $\xi_{2,\ell}$ in $E[X]$ are Galois conjugate to one another. Instead of having a single $\overline{\rho}_q^{\text{sm}}$, we use the proof of [BCGP21b, Theorem 9.2.7] to build $\overline{\rho}_{1,q}^{\text{sm}}$ and $\overline{\rho}_{2,q}^{\text{sm}}$, associated to $\xi_{1,q}$ and $\xi_{2,q}$ respectively, such that

$$(\overline{\rho}_{1,q}^{\text{sm}} \oplus \overline{\rho}_{2,q}^{\text{sm}})|_{G_{\tilde{F}_v}} \cong (\xi_{1,q} \oplus \xi_{2,q})|_{G_{\tilde{F}_v}} \cong A[q]^{\vee}|_{G_{\tilde{F}_v}}$$

and

$$(\overline{\rho}_{1,q}^{\text{sm}} \oplus \overline{\rho}_{2,q}^{\text{sm}})|_{G_{\tilde{F}_w}} \cong \left((\xi_{1,q} \oplus \xi_{2,q})|_{G_{\tilde{F}_w}} \right)^{\text{ss}} \cong \left(A[q]^{\vee}|_{G_{\tilde{F}_w}} \right)^{\text{ss}}.$$

- We can set up our moduli problem to capture abelian fourfolds B for which $B[p]^{\vee} \cong A[p]^{\vee}$ and $B[q]^{\vee} \cong \overline{\rho}_{1,q}^{\text{sm}} \oplus \overline{\rho}_{2,q}^{\text{sm}}$. Previously, in order for A to define a totally real point on our moduli space Y'_A , we needed to prescribe a nontrivial Galois module structure on the geometric endomorphisms of abelian varieties parameterized by Y'_A . Here, because A already has RM by E , we do not need to define a twisted Galois action on the endomorphism algebra when defining Y'_A (i.e. we may take $\rho_{\text{End}} = 1$ when applying Theorem 2.14).

Thus, given the following lemma to ensure that we have enough good primes, the proof goes through in the same way as the proof of Theorem 1.4.

Lemma 5.6. *Let A be a geometrically type I(2) abelian fourfold. The set of good rational primes for A/F has relative density 1 in the set of rational primes which are inert in $E/\mathbb{Q}$, and in particular, have positive relative density in the set of all rational primes.*

Proof. This proof has the same structure as the proof of Lemma 5.2. By Main Theorem 1, $\mathcal{R} = (r_\lambda)_\lambda$ is Schur unobstructed and has coefficients in E . Again, we have $Q_{\mathcal{R},v}(X) := x^4 - a_1(\text{Frob}_v)x^3 + a_2(\text{Frob}_v)x^2 - pa_1(\text{Frob}_v)x + p^2$, and a prime $v|p$ of F fails to be ordinary and p -distinguished ordinary if and only if $v_p(a_2(\text{Frob}_v)) > 0$ or $v_p(a_1(\text{Frob}_v)^2 - 4a_2(\text{Frob}_v)) > 0$. Ultimately, we want to express these conditions using polynomials with rational coefficients, so taking norms, v fails to be ordinary if and only if $p^{f_p} \mid \text{Nm}(a_2(\text{Frob}_v))$ or $p^{f_p} \mid \text{Nm}(a_1(\text{Frob}_v)^2 - 4a_2(\text{Frob}_v))$, where f_p is the inertia degree of p . Again by the Weil bounds, for any embedding $j: \overline{\mathbb{Q}} \hookrightarrow \mathbb{C}$, $|j(a_1(\text{Frob}_v))| \leq 4\sqrt{p}$ and $|j(a_2(\text{Frob}_v) - 2p)| \leq 4p$. Taking norms and rearranging, we obtain the bounds

$$|\text{Nm}(a_2(\text{Frob}_v))| \leq 36p^2 \quad \text{and} \quad |\text{Nm}(a_1(\text{Frob}_v)^2 - 4a_2(\text{Frob}_v))| \leq 1600p^2 \quad (17) \quad \{\text{eq:good-2}\}$$

The Mumford–Tate group of a type I(2) abelian fourfold is $G := \{g \in \text{Res}_{E/\mathbb{Q}} \text{GSp}_{4,E} : \det g \in \mathbb{G}_{m,\mathbb{Q}}\}$. Again we have the Mumford–Tate conjecture in this setting by ???. Choosing any rational prime ℓ inert in $E/\mathbb{Q}$, we define the representation $\rho := \rho_\ell := (r^{\otimes 2} \oplus \wedge^2 r) \otimes \varepsilon^{-1}$, which we think of as valued in $\text{Res}_{E_\lambda/\mathbb{Q}_\ell} \text{GSp}_{22}(E_\lambda)$. Given a polynomial $P(g)$ in the entries of $\text{GSp}_{22}(E_\lambda)$ – and in particular, the tr_1 and tr_2 from Equation (15)) – $\text{Nm} P(g)$ defines a polynomial with rational coefficients in $\text{Res}_{E_\lambda/\mathbb{Q}} \text{GSp}_{22}(E_\lambda) = G_\ell$. If p is inert in $E/\mathbb{Q}$, a prime $v|p$ of F fails the conditions if and only if $\rho(\text{Frob}_v)$ lands in the union Y of the hypersurfaces cut out by the equations

$$\{\text{Nm}(\text{tr}_2(g)) = c : c \in \mathbb{Z} \cap [-36, 36]\} \quad \text{and} \quad \{\text{Nm}(\text{tr}_1(g) - 4\text{tr}_2(g)) = c' : c' \in \mathbb{Z} \cap [-1600, 1600]\}. \quad (18) \quad \{\text{eq:good-2}\}$$

As before, $\rho(\text{Frob}_v)$ avoids Y for a density 1 subset of the primes of F . The remainder of the argument is exactly the same, as we again know the Mumford–Tate conjecture for such fourfolds [MZ95]. $\square$

Remark. The strategy of Lemma 5.6 relies crucially on the existence of infinitely many rational primes with norm $p^{[E:\mathbb{Q}]}$ – by restricting to these primes, we can describe our ordinarity and distinguished ordinarity conditions in Equation (18) using only finitely many equations. This is related to the problem of proving that there are infinitely many ordinary primes for a modular form with coefficients in an extension field $E/\mathbb{Q}$, which is also open for general E .

Remark. In [BCGP21b], a challenging abelian surface is defined to be any surface A which is either geometrically type I(1) (typical, $\mathcal{V}(A) = \mathcal{R}$) or geometrically type I(2) with $\mathcal{V}(\cong \text{Ind}_K^F \mathcal{R})$ for a two-dimensional compatible system of G_K -representations $\mathcal{R}$. The former are challenging because $\mathcal{R}$ is irregular, and the latter are challenging because K need not be totally real. Our notion of a challenging type I(2) abelian fourfold is a generalization of the latter. [BCGP21b] deal with challenging type I(2) abelian surfaces by treating them alongside the case of typical abelian surfaces – they Moret-Bailly to find a point B on an appropriately twisted Siegel threefold with p -torsion related to A and automorphic q -torsion. We cannot do the same in our case because B would generically be a typical abelian fourfold, and we do not have a modularity lifting theorem in this setting.

5.3 Modularity of atypical abelian fourfolds beyond Main Theorem 2

If A is geometrically type III, $\text{End}_{\overline{F}}(A)$ is a quaternion algebra with center $\mathbb{Q}$ [MZ95]. By Main Theorem 1, $\mathcal{V} \cong \mathcal{R} \otimes \mathcal{A}$, where $\mathcal{R}$ is of orthogonal type. Potential modularity of four-dimensional geometric orthogonal representations with Hodge–Tate weights $(0, 0, 1, 1)$ is open, so we cannot immediately deduce potential modularity of geometrically type III abelian fourfolds. We plan to address this problem in future work.

The remaining atypical case is that of geometrically type IV(1,1) abelian fourfolds, which includes the famous “Weil type” abelian fourfolds.

5.4 Even symplectic representations appearing in geometry

subsec:even

To the best of this author’s knowledge, all known modularity theorems for symplectic (resp. orthogonal) Galois representations assume that the representations in question are odd (resp. even). Not every symplectic (resp. orthogonal) compatible system appearing in geometry is odd (resp. even). Indeed, it is not hard – for example by tensoring an elliptic curve by an odd Artin representation – to produce even symplectic compatible systems “appearing in” geometry. However, the methods of this paper suggests a more refined question – is the strongly irreducible system $\mathcal{R}$ appearing in Main Theorem 1 odd if and only if it is symplectic (resp. even if and only if it is orthogonal)? For $g = 1$ and $g = 2$, $\mathcal{R}$ is always symplectic or unitary, and in the symplectic cases it follows from Lemma 4.4 and Main Theorem 1 that $\mathcal{R}$ is odd. When $g = 4$ and A is geometrically type III (i.e. A has geometric endomorphisms by a definite quaternion algebra over $\mathbb{Q}$), one can deduce (as we learned from Frank Calegari) that the geometric endomorphisms of A cannot be defined over the reals. Equivalently, writing $\mathcal{V} \cong \text{Des}_{\mathbb{Q}}(\mathcal{R} \otimes \mathcal{A})$ as in Main Theorem 1, the two-dimensional symplectic Artin compatible system $\mathcal{A}$ is odd, and hence the four-dimensional (orthogonal, by Lemma 3.6) compatible system $\mathcal{R}$ is even. However, we will show that there exist abelian varieties such that $\mathcal{R}$ is symplectic and even! To prove a modularity result in this setting, we need some way to attach a Siegel Maass form to such representations, which seems far out of reach at present.

The following theorem and its proof, which the author learned from Frank Calegari, were the motivation for the general twisted moduli construction given in Section 2.6 We are grateful to him for allowing us to include it in this article.

lem:even-gsp4

Lemma 5.7 (F. Calegari, private communication). *Let $D/\mathbb{Q}$ be an indefinite quaternion algebra equipped with a positive involution $\dagger$. There exists a g -dimensional abelian variety A over a totally real field F such that $\text{End}_F^0(A) = D$ and $\text{End}_F^0(A) = E'$, where E' is the distinguished quadratic subfield of $(D, \dagger)$ as defined in Lemma 4.2, if and only if $g \equiv 0 \pmod{4}$.*

Remark. If $\text{End}_F^0(A) = E'$ for some E'/E , and E'/E is CM, then $\dagger$ must restrict to complex conjugation on E'/E by the Albert classification. Because D/E is indefinite, the subspace of D on which $\dagger$ acts with eigenvalue -1 is 1-dimensional over E , so E' must be the distinguished quadratic subfield of $(D, \dagger)$.

Proof. Choose i such that $E' = \mathbb{Q}(i) \subset D$ and $i^2 = a$ is a totally negative element of E . Choose $j \in D$ such that $ij = -ji$ and j^2 is a totally positive element b of E . Such a j exists because D is indefinite. Let c be the nontrivial element of $\text{Gal}(\mathbb{C}/\mathbb{R})$. We apply Theorem 2.14 with $F' = \mathbb{Q}$, and L' an arbitrary imaginary quadratic field. We define $\rho_D(c) = (d \mapsto idi^{-1})$, and take $S := \emptyset$ (no level structure).

The idea of this construction is that a real point on $Y'/\mathbb{R}$ will correspond to a polarized abelian variety $A/\mathbb{R}$ with $\text{End}_{\mathbb{R}}^0(A) = E'$ (the subalgebra of D fixed by $\rho_D(c)$) and $\text{End}_{\mathbb{C}}^0(A) = D$. We

will show that $Y'(\mathbb{R})$ is nonempty if and only if $g \equiv 0 \pmod{4}$, in which case it follows from the Moret-Bailly theorem that Y' has a totally real point.

If g is odd, then A cannot have geometric endomorphisms by a quaternion algebra by the Albert classification (e.g. because if $H^1(A(\mathbb{C}), \mathbb{Q})$ is a free D -module, then 4 must divide $2g$).

Suppose that $g \equiv 2 \pmod{4}$, and $(A/\mathbb{R}, \lambda, \iota, \eta)$ is the quadruple associated to a point in $Y'(\mathbb{R})$. The Lie algebra $\text{Lie}(A)$ is a g -dimensional real vector space, and $\text{Lie}(A_{\mathbb{C}}) = \text{Lie}(A) \otimes_{\mathbb{R}} \mathbb{C}$. Because A is defined over $\mathbb{R}$, there is a descent datum $f'_c: A_{\mathbb{C}} \rightarrow A_{\mathbb{C}}$ lying over complex conjugation on $\text{Spec } \mathbb{C}$ such that $(f'_c)^2 = 1$. The induced map on $\text{Lie}(A_{\mathbb{C}})$, which we denote by γ , is a $\mathbb{C}$ -antilinear involution from $\text{Lie}(A_{\mathbb{C}})$ to itself.

Let $\iota: D \hookrightarrow \text{End}^0(A_{\mathbb{C}})$. Because $\iota(i)$ is defined over $\mathbb{R}$ by the construction of Y' , the endomorphism $\iota(i)$ acts $\mathbb{R}$ -linearly on $\text{Lie}(A)$, and $\iota(j)$ acts $\mathbb{C}$ -linearly on $\text{Lie}(A_{\mathbb{C}})$. Because $\iota(i)$ is defined over $\mathbb{R}$ and γ acts as conjugation by i on D , we have the relations

$$\iota(i)^2 = -a, \iota(j)^2 = b, \gamma^2 = 1, \iota(i)\iota(j) = -\iota(j)\iota(i), \iota(i)\gamma = \gamma\iota(i), \quad \text{and } \gamma\iota(j) = -\iota(j)\gamma.$$

Let $W \subset \text{Lie}(A_{\mathbb{C}}) = \text{Lie}(A) \otimes_{\mathbb{R}} \mathbb{C}$ be the $\sqrt{-a}$ -eigenspace of $\iota(i)$. Because $\iota(j)$ and γ anticommute with $\iota(i)$, the product $\iota(j)\gamma$ acts on W . This product is $\mathbb{C}$ -antilinear, and hence can be written as $\circ \alpha$, where $\circ$ denotes complex conjugation (on the second factor in $\text{Lie}(A) \otimes_{\mathbb{R}} \mathbb{C}$) and α is a $\mathbb{C}$ -linear map on the $\frac{g}{2}$ -dimensional vector space W . Then,

$$-bI = (\iota(j)\gamma)^2 = (\circ \alpha)^2 = \bar{\alpha}\alpha,$$

so $|\det(\alpha)|^2 = (-b)^{g/2} < 0$, but this is a contradiction.

Now suppose that $g \equiv 0 \pmod{4}$. Let $\mathcal{O}$ be a $\mathbb{Z}$ -lattice in D fixed by $\dagger$, and define $\Lambda := \mathcal{O} \oplus \mathcal{O}$. Let $\phi: \Lambda \times \Lambda \rightarrow \mathcal{O}$ be the alternating $\mathcal{O}$ -skew Hermitian pairing given by the matrix $J = \begin{pmatrix} & 1 \\ -1 & \end{pmatrix}$. Then, defining $h: \text{Res}_{\mathbb{C}/\mathbb{R}} \mathbb{G}_m \rightarrow \text{GL}(\Lambda \otimes_{\mathbb{Z}} \mathbb{R})$ by $h(i) = J$, Λ is a polarized weight 1 $\mathcal{O}$ -Hodge structure with polarization $\text{tr}_{\mathcal{O}/\mathbb{Z}} \phi$, and hence corresponds to a complex abelian variety $A/\mathbb{C}$. A descent datum on $A/\mathbb{C}$ corresponds to an involution $\delta: H^1(A(\mathbb{C}), \mathbb{Z}) \rightarrow H^1(A(\mathbb{C}), \mathbb{Z})$ such that $\delta^2 = 1$ and $\delta J = -J\delta$. Let $\mathcal{O}_{E'} := \mathcal{O} \cap E'$, and fix an E -linear embedding $\mathcal{O} \hookrightarrow M_2(\mathcal{O}_{E'})$ given by

$$1 \mapsto \begin{pmatrix} 1 & \\ & 1 \end{pmatrix}, i \mapsto \begin{pmatrix} i & \\ & i \end{pmatrix}, j \mapsto \begin{pmatrix} \alpha & \beta \\ -\beta & -\alpha \end{pmatrix}, ij \mapsto \begin{pmatrix} -i\beta & -i\alpha \\ -i\alpha & -i\beta \end{pmatrix}. \quad (19) \quad \text{\color{blue}\{eq:embed\}}$$

In order for the descended abelian variety A to give rise to a real point on Y' , we would like $\delta \in M_4(\mathcal{O}_{E'}) \setminus M_2(\mathcal{O})$, along with the existing conditions of $\delta J = -J\delta$ and $\delta^2 = 1$. The latter two conditions force $\delta = \begin{pmatrix} A & B \\ B & -A \end{pmatrix}$ for matrices A and B such that $A^T = A$, $B^T = B$, and $A^2 + B^2 = I$.

For this, we take $A = 0$ and $B = \begin{pmatrix} & 1 \\ 1 & \end{pmatrix}$ – note that any symmetric matrix in $M_2(\mathcal{O}_{E'})$ in the image of Equation (19) has totally imaginary top-right and bottom-left entries, so δ is indeed not in $M_2(\mathcal{O})$. $\square$

Main Theorem 3. *There exist type II abelian fourfolds A/F , F a totally real field, for which the associated four-dimensional compatible system $\mathcal{R}$ is even and symplectic-type. In particular, there exists a representation $\xi: G_F \rightarrow \text{GSp}_4(\overline{\mathbb{Q}}_\ell)$ such that ξ is geometric, has Zariski dense image, and is totally even.*

Proof. Let D be an indefinite quaternion algebra with positive involution $\dagger$ and let F be a totally real field and A/F an abelian fourfold as given by Main Theorem 3. By Main Theorem 1, for an abelian fourfold we have $\mathcal{V} \cong \mathcal{R} \otimes \mathcal{A}$. Because $\text{End}_F^0(A)$ is a CM field, $\mathcal{A}$ is cyclic, and by Equation (12) and Lemma 4.3, the similitude character is η . Because L is totally imaginary, η is odd. $\square$

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